Supplementary Material

Well-posedness, Stability, and Bifurcation in an Acid-Mediated Tumour Invasion Model with Interspecific Competition

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This document holds the full proofs outlined in the main text and the technical material behind its numerical study. Everything here is numbered S1, S2, ... (sections, statements, equations, figures); a reference without the letter S points to the main manuscript.

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S1 Full derivation of the nondimensional system

Starting from the dimensional system

$$\begin{aligned}
\partial_{\tilde{t}}\tilde{N} &= \tilde{r}_1\tilde{N}\Big(1-\tilde{N}/\tilde{K}_N\Big) - \tilde{d}_1\tilde{H}\tilde{N} + \tilde{D}_1\nabla^2_{\tilde{x}}\tilde{N},\\
\partial_{\tilde{t}}\tilde{T} &= \tilde{r}_2\tilde{T}\Big(1-\tilde{T}/\tilde{K}_T - \tilde{\alpha}\tilde{N}/\tilde{K}_N\Big) + \tilde{D}_2\nabla^2_{\tilde{x}}\tilde{T},\\
\partial_{\tilde{t}}\tilde{H} &= \tilde{r}_3\tilde{T} - \tilde{d}_3\tilde{H} + \tilde{D}_3\nabla^2_{\tilde{x}}\tilde{H},
\end{aligned} \tag{S1}$$

Table S1: Dimensional values of the model (S1), from Martin et al. (2010, Table 1). The unit of $\tilde{d}_1$, printed there as cells $\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, is read as $\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, the unit that makes $\tilde{d}_1\tilde{H}\tilde{N}$ a rate of change of a cell density. From `critical_radius.py`.

	Meaning	**Value**	**Unit**
$\tilde{r}_1$	normal-cell growth rate	$2 \cdot 10^{-6}$	s^{-1}
$\tilde{r}_2$	tumour-cell growth rate	$3.2 \cdot 10^{-6}$	s^{-1}
$\tilde{K}_N, \tilde{K}_T$	carrying capacities	$5 \cdot 10^{8}$	cells cm^{-3}
$\tilde{D}_2$	tumour-cell diffusivity	$2 \cdot 10^{-10}$	$\mathrm{cm}^2\ \mathrm{s}^{-1}$
$\tilde{D}_3$	effective acid diffusivity	$5 \cdot 10^{-6}$	$\mathrm{cm}^2\ \mathrm{s}^{-1}$
$\tilde{r}_3$	H^+ production rate	$1.1 \cdot 10^{-17}$	$\mathrm{M\ cm}^3\ \mathrm{s}^{-1}\ \mathrm{cell}^{-1}$
$\tilde{d}_3$	acid clearance rate	$1.1 \cdot 10^{-4}$ – 0.024	s^{-1}
$\tilde{d}_1$	acid-induced death rate	0 – 10	$\mathrm{M}^{-1}\ \mathrm{s}^{-1}$

we introduce the rescaled variables

$$N = \tilde{N}/\tilde{K}_N, \quad T = \tilde{T}/\tilde{K}_T, \quad H = \tilde{H}/\tilde{H}_0, \quad t = \tilde{r}_1\,\tilde{t}, \quad x = \tilde{x}/\tilde{X}_0, \qquad \tilde{X}_0 = \sqrt{\tilde{D}_3/\tilde{r}_1},$$

where the reference acid concentration $\tilde{H}_0 > 0$ is left free at this stage. Dividing the first equation of (S1) by $\tilde{r}_1\tilde{K}_N$ and using $\nabla^2_{\tilde{x}} = \tilde{X}_0^{-2}\nabla^2$ gives $\partial_t N = N(1-N) - \delta HN + \delta_1\nabla^2 N$ with $\delta = \tilde{d}_1\tilde{H}_0/\tilde{r}_1$ and $\delta_1 = \tilde{D}_1/\tilde{D}_3$. Dividing the second by $\tilde{r}_1\tilde{K}_T$ gives $\partial_t T = \beta\,T(1-T-\alpha N) + \delta_2\nabla^2 T$ with $\beta = \tilde{r}_2/\tilde{r}_1$ and $\delta_2 = \tilde{D}_2/\tilde{D}_3$, and dividing the third by $\tilde{r}_1\tilde{H}_0$ gives $\partial_t H = \gamma T - \omega H + \nabla^2 H$ with $\gamma = \tilde{r}_3\tilde{K}_T/(\tilde{r}_1\tilde{H}_0)$ and $\omega = \tilde{d}_3/\tilde{r}_1$. The threshold $\delta_c = \omega/\gamma = \tilde{d}_3/(\tilde{r}_3\tilde{K}_T/\tilde{H}_0)$ is the ratio of acid clearance to acid supply per unit reference concentration.

Remark S1.1 (Choice of $\tilde{H}_0$; complement to Remark 2.1). Under $\tilde{H}_0 \mapsto \lambda\tilde{H}_0$, $H \mapsto H/\lambda$, $\gamma \mapsto \gamma/\lambda$, $\delta \mapsto \lambda\delta$ and ω is unchanged, so only $\delta\gamma$ enters the results. Keeping $\tilde{H}_0$ free leaves visible the roles of $\tilde{r}_3$ and $\tilde{d}_3$, which distinguish the phenotypes of Table 1.

S1.1 Dimensional values

Table S1 collects the dimensional values compiled by Martin et al. (2010, Table 1) from Gatenby and Gawlinski (1996) and Gatenby et al. (2006); they are typed once, in `critical_radius.py`, which writes this table. They imply $\beta = 1.6$, against 1.5 in our simulations, and $\delta_2 = \tilde{D}_2/\tilde{D}_3 = 4 \cdot 10^{-5}$, against $5 \cdot 10^{-3}$ (main text, Section 4.1). The clearance rate is the least constrained entry: it spans more than two decades, and with it $\omega = \tilde{d}_3/\tilde{r}_1$ runs from 55 to $1.2 \cdot 10^4$. A tumour-wide drop of the extracellular pH by Δ units from 7.4 fixes the excess acid at carrying capacity, $\tilde{H}_{\mathrm{sat}} = \tilde{r}_3\tilde{K}_T/\tilde{d}_3 = 10^{-7.4}(10^{\Delta} - 1)$ M, hence the clearance time; for $\Delta = 0.5$ and 1 it gives $1/\tilde{d}_3 = 16$ s and 65 s, which bracket the fastest tabulated value, 42 s, and correspond to $\omega = 7.7 \cdot 10^3$ to $3.2 \cdot 10^4$; conversely, with the same production, clearance times of 15 min and 2.5 h would lower the pH to 5.3 and 4.3. The toxicity ratio (4.1), $\tilde{d}_1\tilde{r}_3\tilde{K}_T/(\tilde{r}_1\tilde{d}_3)$, ranges from 0 to 1.1 at the fastest clearance and to 250 at the slowest, so that the values $\delta/\delta_c = 0.31$ and 1.56 of Table 1 lie inside the documented range without being fitted to it. These are orders of magnitude: the production rate $\tilde{r}_3$ was fitted within a model that also resolves the vasculature (Martin et al. 2010), and the range of $\tilde{d}_1$ is a range from which values are selected, not a measurement.

S2 Proofs of the well-posedness results

This section proves the results of Sections 3.1 and 3.2.

S2.1 The energy class

This subsection supports Remark 2.3: the pointwise bound of Definition 2.2 is not needed to give the weak equation a meaning, nor for uniqueness. For $d \le 3$, the Gagliardo–Nirenberg inequality $\|v\|_{L^{12/5}} \le C\,\|v\|_{L^2}^{3/4}\|v\|_{H^1}^{1/4}$ bounds every quadratic term of F by $\|u_iu_j\|_{L^{6/5}} \le C\,\|u\|_{\mathbf{H}}^{3/2}\|u\|_{\mathbf{V}}^{1/2}$, so that an energy-class $u \in L^\infty(0,T_{\mathrm{end}};\mathbf{H}) \cap L^2(0,T_{\mathrm{end}};\mathbf{V})$ already has $F(u) \in L^2(0,T_{\mathrm{end}};L^{6/5}) \subset L^2(0,T_{\mathrm{end}};\mathbf{V}')$, since $H^1 \hookrightarrow L^6$.

Uniqueness holds among energy-class solutions as well. If $w = u - v$, the reaction terms being at most quadratic, $\langle F(u) - F(v), w\rangle$ is bounded by $C\int_\Omega(1+|u|+|v|)|w|^2$, and $\int_\Omega |u||w|^2 \le \|u\|_{L^3}\|w\|_{L^3}^2 \le C\|u\|_{L^3}\|w\|_{L^2}\|w\|_{H^1}$; Young's inequality absorbs $\|\nabla w\|_{L^2}^2$ into the dissipation and leaves the Gronwall weight $C(1+\|u\|_{L^3}^2+\|v\|_{L^3}^2)$, integrable in time because $\|u\|_{L^3}^2 \le C\|u\|_{\mathbf{H}}\|u\|_{\mathbf{V}}$ and $u \in L^\infty(\mathbf{H}) \cap L^2(\mathbf{V})$.

In the bounded class used in the main text, F is Lipschitz on $\mathcal{R}$ and the uniqueness argument of Step 6 of the proof of Theorem 3.4 needs no interpolation. With u bounded, $F(u) \in L^\infty(\mathbf{H})$, and the Lions–Magenes embedding $L^2(0,T_{\mathrm{end}};\mathbf{V}) \cap H^1(0,T_{\mathrm{end}};\mathbf{V}') \hookrightarrow C([0,T_{\mathrm{end}}];\mathbf{H})$ (Lions and Magenes 1972) gives $u \in C([0,T_{\mathrm{end}}];\mathbf{H})$, so the initial condition is well defined.

S2.2 Proof of Theorem 3.1

Proof of Theorem 3.1. On $\{N,T,H \ge 0\}$ the off-diagonal derivatives of F are

$$\partial_H F_1 = -\delta N \le 0, \qquad \partial_N F_2 = -\alpha\beta T \le 0, \qquad \partial_T F_3 = \gamma > 0,$$

and $\partial_T F_1 = \partial_H F_2 = \partial_N F_3 \equiv 0$, so F is mixed quasimonotone and the existence–comparison theorem for such systems (Pao 1992, Ch. 8) applies to a pair $(\bar u, \underline u)$ of coupled upper and lower solutions: in the inequality for $\bar u$, each F_i is evaluated at $\bar u$ in the variables in which it increases and at $\underline u$ in those in which it decreases, its own variable at $\bar u$; the roles are exchanged for $\underline u$. For the constants $(\bar u, \underline u) = ((1,1,M_0),(0,0,0))$, ∂_t and Δ vanish and the inequalities reduce to sign conditions on F:

$$F_1(\bar u_1, \cdot\,, \underline u_3) = 1\cdot(1-1-0) = 0, \qquad F_2(\underline u_1, \bar u_2, \cdot) = \beta\,(1-1-0) = 0,$$

$$F_3(\cdot\,, \bar u_2, \bar u_3) = \gamma - \omega M_0 \le 0,$$

all three non-positive, the last since $M_0 \ge \gamma/\omega$, and

$$F_1(\underline u_1, \cdot\,, \bar u_3) = 0\cdot(1-0-\delta M_0) = 0, \qquad F_2(\bar u_1, \underline u_2, \cdot) = 0, \qquad F_3(\cdot\,, \underline u_2, \underline u_3) = 0.$$

Constants satisfy the Neumann condition and $\underline u \le u_0 \le \bar u$, so on the $C^{2,\nu}$ domain the existence–comparison theorem yields a classical solution v with datum u_0 and $\underline u \le v \le \bar u$, that is, $v \in \mathcal{R}$.

That theorem gives uniqueness only in $\langle \underline u, \bar u\rangle$; we prove it in the whole classical class. Let u, v be classical solutions with the same datum and $T' < T_{\mathrm{end}}$. On $\bar\Omega \times [0,T']$ both are bounded by some R, F is L_R-Lipschitz on $\{|w| \le R\}$, and $w = u - v$ satisfies $\partial_\nu w = 0$, so the divergence theorem gives

$$\tfrac12 \tfrac{d}{dt}\|w(t)\|_{\mathbf{H}}^2 + d_*\|\nabla w(t)\|_{\mathbf{H}}^2 \;\le\; \int_\Omega \big(F(u) - F(v)\big)\cdot w\,dx \;\le\; L_R\,\|w(t)\|_{\mathbf{H}}^2$$

for $0 < t < T'$. Gronwall on $[\varepsilon, t]$ and $\|w(\varepsilon)\|_{\mathbf{H}} \to 0$ (by continuity, $w(0) = 0$) give $w \equiv 0$ on $[0,T']$; hence $u = v \in \mathcal{R}$.

Geometrically, invariance (Chueh et al. 1977) requires the outward normals $\pm e_i$ of $\mathcal{R}$ to be left eigenvectors of D, true because D is diagonal (cross-diffusion would break it), and F to point inward: $F_1(0,T,H) = F_2(N,0,H) = 0$ and $F_3(N,T,0) = \gamma T \ge 0$ on the lower faces; $F_2(N,1,H) = -\alpha\beta N \le 0$, $F_3(N,T,M_0) = \gamma T - \omega M_0 \le 0$ and $F_1(1,T,H) = -\delta H \le 0$ on the upper ones, valid once $N \ge 0$, $T \le 1$ and $H \ge 0$ respectively. Thus $H \le M_0$ needs $T \le 1$, while $N \le 1$ needs only $H \ge 0$. □

S2.3 Proof of Lemma 3.2

Proof of Lemma 3.2. We redo the sign computations of Theorem 3.1 for bounded weak solutions, comparison being replaced by Stampacchia truncation; boundedness is essential. By Definition 2.2, $u \in L^\infty(\Omega \times (0,T_{\mathrm{end}}))$, so

$$R_0 := \|u\|_{L^\infty(\Omega\times(0,T_{\mathrm{end}}))} < \infty, \tag{S2}$$

a constant of the given solution. All constants below depend on R_0 and the parameters only, hence no circularity: the quadratic reaction is Lipschitz on $\{|u| \le R_0\}$, not on the energy class. Each truncation used as test function lies in $\mathbf{V}$ (chain rule, Evans 2010) and vanishes at $t = 0$ as $u_0 \in \mathcal{R}$; the time derivative of its squared norm is given by Alt and Luckhaus (1983, Lemma 1.5), and the non-negative diffusive term is discarded.

Step 1 (positivity). Test the N-equation with $v = -N^-$, $N^- = \max(-N, 0)$. As $F_1(0, T, H) = 0$, on $\{N < 0\}$ $F_1(N, T, H)\,(-N^-) = (N^-)^2(1 - N - \delta H)$, with $|1 - N - \delta H| \le 1 + R_0 + \delta R_0$ by (S2). Hence

$$\tfrac{1}{2}\tfrac{d}{dt}\|N^-\|_{L^2}^2 + \delta_1\|\nabla N^-\|_{L^2}^2 \;\le\; C\,\|N^-\|_{L^2}^2, \qquad C = 1 + (1+\delta)R_0,$$

finite by boundedness, not by the bound being proved. Gronwall and $N^-(0) = 0 \Rightarrow N^- \equiv 0$, i.e. $N \ge 0$ a.e. Likewise for T, using $F_2(N, 0, H) = 0$ and $|1 - T - \alpha N| \le 1 + (1+\alpha)R_0$: $T \ge 0$ a.e., with no sign information on N needed. For H, test with $-H^-$: on $\{H < 0\}$ the reaction gives $(\gamma T - \omega H)\,H = \gamma T H - \omega H^2 \le 0$ as $T \ge 0$, so $H^- \equiv 0$.

Step 2 (upper bound on T). Test the T-equation with $v = (T-1)^+$:

$$\tfrac{1}{2}\tfrac{d}{dt}\|(T-1)^+\|_{L^2}^2 + \delta_2\|\nabla(T-1)^+\|_{L^2}^2 = \int_\Omega \beta\, T\,(1 - T - \alpha N)\,(T-1)^+\,dx.$$

On $\{T > 1\}$, $T > 0$ and the bracket is $-(T-1) - \alpha N \le 0$ ($N \ge 0$ by Step 1), so the right-hand side is ≤ 0; with $(T-1)^+(0) = 0$, $T \le 1$ a.e.

Step 3 (upper bound on H). Test the H-equation with $v = (H - M_0)^+$:

$$\tfrac{1}{2}\tfrac{d}{dt}\|(H-M_0)^+\|_{L^2}^2 + \|\nabla(H-M_0)^+\|_{L^2}^2 = \int_\Omega (\gamma T - \omega H)\,(H-M_0)^+\,dx.$$

On $\{H > M_0\}$, $T \le 1$ (Step 2) and the definition of M_0 give $\gamma T - \omega H \le \gamma - \omega M_0 \le 0$; hence $H \le M_0$ a.e.

Step 4 (upper bound on N). Test the N-equation with $v = (N-1)^+$; on $\{N > 1\}$, $N(1 - N - \delta H) = N\big(-(N-1) - \delta H\big) \le 0$ as $H \ge 0$ (Step 1), so $N \le 1$ a.e.

Steps 1–4 use respectively nothing, Step 1, Step 2, Step 1: no circularity, and $u(x,t) \in \mathcal{R}$ a.e. □

S2.4 Proof of Theorem 3.4

Proof of Theorem 3.4. Galerkin directly on (2.2) fails: F is only *locally* Lipschitz on $\mathbb{R}^3$, so Galerkin solutions may blow up, and the L^∞ bounds of Lemma 3.2 that would exclude it are unavailable (the orthogonal projection P_m onto V_m does not preserve L^∞: neither $P_m u_0$ nor the iterates need lie in $\mathcal{R}$); arguing on the original system is circular. Following Pierre (2010) and Smoller (1994), we regularise, then identify.

Step 1 (a globally Lipschitz regularisation). Fix $\chi \in C^\infty(\mathbb{R})$ with $\chi(s) = s$ on $[-1, 2]$, $\chi = -2$ on $(-\infty, -3]$, $\chi = 3$ on $[4, \infty)$, $0 \le \chi' \le 1$. One is $\chi(s) = \int_0^s \rho$, where $\rho = 1$ on $[-1, 2]$, $\rho(s) = \varphi((s-2)/2)$ for $s \ge 2$ and $\varphi((-1-s)/2)$ for $s \le -1$, with $\varphi(t) = \psi(1-t)/(\psi(t) + \psi(1-t))$ and $\psi(t) = e^{-1/t}\mathbf{1}_{t>0}$: φ is smooth, non-increasing, 1 on $(-\infty, 0]$, 0 on $[1, \infty)$, and $\varphi(t) + \varphi(1-t) = 1$ gives $\int_0^1 \varphi = \frac{1}{2}$, hence $\chi(4) = 2 + 2\int_0^1 \varphi = 3$ and likewise $\chi(-3) = -2$. Put $M_1 = \max(2, M_0 + 1)$ and $\chi_M(s) = M_1\,\chi(s/M_1)$: $\chi_M(s) = s$ for $|s| \le M_1$, $|\chi_M| \le 3M_1$ and $0 \le \chi_M' \le 1$ on $\mathbb{R}$. Define

$$\widehat{F}(N, T, H) := F\big(\chi_M(N),\, \chi_M(T),\, \chi_M(H)\big).$$

Then $\widehat{F}$ is bounded and globally Lipschitz on $\mathbb{R}^3$ (constant $L_{\widehat{F}}$ depending only on $(\alpha, \beta, \gamma, \delta, \omega, M_1)$), and $\widehat{F} = F$ on $\mathcal{R} \subset [-M_1, M_1]^3$, where $\chi_M = \mathrm{id}$.

Step 2 (Galerkin on the regularised system). Let $(w^\ell)_{\ell \ge 1}$ be an L^2-orthonormal basis of Neumann eigenfunctions of $-\Delta$ on Ω, $V_m = \mathrm{span}\{w^1, \dots, w^m\}$; seek $u^m(t) \in V_m^3$ with

$$(\partial_t u^m, v) + a(u^m, v) = (\widehat{F}(u^m), v) \quad \forall v \in V_m^3, \qquad u^m(0) = P_m u_0. \tag{S3}$$

By Cauchy–Lipschitz ($\widehat{F}$ globally Lipschitz, bounded), it has a unique solution on $[0, T_{\mathrm{end}}]$.

Step 3 (a priori bounds, uniform in m). Take $v = u^m$ in (S3); with $\|\widehat{F}(w)\|_{\mathbf{H}} \le C_F := \|\widehat{F}\|_{L^\infty}|\Omega|^{1/2}$ $\forall w$ and $a(u^m, u^m) \ge d_*\|\nabla u^m\|^2_{\mathbf{H}}$,

$$\tfrac{1}{2}\tfrac{d}{dt}\|u^m\|^2_{\mathbf{H}} + d_*\|\nabla u^m\|^2_{\mathbf{H}} \;\le\; C_F\,\|u^m\|_{\mathbf{H}} \;\le\; \tfrac{1}{2}\|u^m\|^2_{\mathbf{H}} + \tfrac{1}{2}C_F^2.$$

Gronwall yields (3.1) uniformly in m ($\|P_m u_0\|_{\mathbf{H}} \le \|u_0\|_{\mathbf{H}}$), and (S3) then bounds $(u^m)'$ uniformly in $L^2(0,T_{\mathrm{end}};\mathbf{V}')$.

Step 4 (compactness and passage to the limit). By Step 3 and Aubin–Lions (Aubin 1963; Simon 1987), a subsequence (not relabelled) and some $\widehat{u} \in L^2(\mathbf{V}) \cap L^\infty(\mathbf{H})$, $\widehat{u}' \in L^2(\mathbf{V}')$, satisfy $u^m \rightharpoonup \widehat{u}$ in $L^2(\mathbf{V})$, $u^m \to \widehat{u}$ in $L^2(\mathbf{H})$ and a.e. in $\Omega \times (0,T_{\mathrm{end}})$. By dominated convergence ($\widehat{F}$ continuous, bounded), $\widehat{F}(u^m) \to \widehat{F}(\widehat{u})$ in $L^2(\mathbf{H})$, and the limit satisfies

$$\tfrac{d}{dt}(\widehat{u}, v) + a(\widehat{u}, v) = (\widehat{F}(\widehat{u}), v) \qquad \forall v \in \mathbf{V},\ \text{a.e. } t, \qquad \widehat{u}(0) = u_0. \tag{S4}$$

Step 5 (the regularised solution lies in $\mathcal{R}$, and solves the original system). With $g := \widehat{F}(\widehat{u}) \in L^\infty(\Omega \times (0,T_{\mathrm{end}}))$ ($\widehat{F}$ bounded), (S4) says that $\widehat{u} \in L^2(\mathbf{V}) \cap H^1(\mathbf{V}')$ weakly solves the *linear* problem $v' = D\Delta v + g$, homogeneous Neumann data, $v(0) = u_0$, which has at most one weak solution in this class: the difference v of two has $v(0) = 0$ and $\tfrac{1}{2}\tfrac{d}{dt}\|v\|^2_{\mathbf{H}} + a(v,v) = 0$ a.e. t (licit: $v \in L^2(\mathbf{V})$, $v' \in L^2(\mathbf{V}') \Rightarrow t \mapsto \|v(t)\|^2_{\mathbf{H}}$ is absolutely continuous with derivative $2\langle v', v\rangle$, Lions and Magenes 1972), so $\|v(t)\|_{\mathbf{H}}$ is non-increasing and vanishes. Let $\{e^{\tau D\Delta_N}\}_{\tau\ge 0}$ be the analytic semigroup on $\mathbf{H}$ generated by the Neumann realisation $D\Delta_N$ of $D\Delta$ (self-adjoint, associated with a), acting as $e^{\tau d_i \Delta_N}$ on the i-th component (D diagonal). The mild solution

$$v(t) \;=\; e^{tD\Delta_N}u_0 \;+\; \int_0^t e^{(t-s)D\Delta_N}\, g(s)\, ds$$

is, by maximal regularity of the form method ($u_0 \in \mathbf{H}$, $g \in L^2(0,T_{\mathrm{end}};\mathbf{H})$) (Lions and Magenes 1972), in $L^2(\mathbf{V}) \cap H^1(\mathbf{V}')$ with $\tfrac{d}{dt}(v,\varphi) + a(v,\varphi) = (g,\varphi)$ $\forall \varphi \in \mathbf{V}$, a.e. t; by uniqueness, $\widehat{u} = v$. Each block $e^{\tau d_i\Delta_N}$, generated by a Dirichlet form on a bounded Lipschitz domain, is positivity preserving and fixes constants, hence is an $L^\infty(\Omega)$ contraction (Ouhabaz 2005), and Duhamel gives

$$\|\widehat{u}(t)\|_{L^\infty} \;\le\; \|u_0\|_{L^\infty} + \int_0^t \|g(s)\|_{L^\infty}\, ds \;\le\; \|u_0\|_{L^\infty} + T_{\mathrm{end}}\,\|\widehat{F}\|_{L^\infty},$$

so $\widehat{u} \in L^\infty(\Omega \times (0,T_{\mathrm{end}}))$ is a bounded weak solution (Definition 2.2), to which the Stampacchia chain of Lemma 3.2 applies if its four sign computations survive $F \to \widehat{F}$ ($\ne F$ off $\mathcal{R}$). Recall χ_M is non-decreasing, $\chi_M(0) = 0$, $|\chi_M| \le 3M_1$, so $0 \le \chi_M(s)\,s \le s^2$. Testing with $-N^-$, the reaction contributes $\int_{\{N<0\}} \chi_M(N)\,N\,\big(1 - \chi_M(N) - \delta\chi_M(H)\big) \le \big(1 + 3M_1(1+\delta)\big)\|N^-\|^2_{L^2}$, and Gronwall gives $N \ge 0$ with *no* appeal to $\widehat{F} = F$ (boundedness of χ_M removes the cubic term); likewise, with bracket bound $1 + 3M_1(1+\alpha)$, $T \ge 0$. Testing with $-H^-$, $\int_{\{H<0\}}\big(\gamma\chi_M(T) - \omega\chi_M(H)\big)H \le 0$, as $\chi_M(T) \ge 0$ ($T \ge 0$) and $\chi_M(H)H \ge 0$; hence $H \ge 0$. On $\{T > 1\}$: $N \ge 0 \Rightarrow \chi_M(N) \ge 0$ ($= N$ if $N \le M_1$), and $\chi_M(T) \ge \chi_M(1) = 1$; hence $\widehat{F}_2 \le 0$. On $\{H > M_0\}$: $T \in [0,1] \subset [-M_1, M_1] \Rightarrow \chi_M(T) = T \le 1$, and $\chi_M(H) \ge \chi_M(M_0) = M_0$ ($M_0 \le M_1$); hence $\widehat{F}_3 \le \gamma - \omega M_0 \le 0$. On $\{N > 1\}$: $H \in [0, M_0] \subset [-M_1, M_1] \Rightarrow \chi_M(H) = H \ge 0$, and $\chi_M(N) \ge 1$; hence $\widehat{F}_1 \le 0$. The chain closes as in Lemma 3.2: $\widehat{u}(x,t) \in \mathcal{R}$ a.e.

As $\widehat{F} = F$ *as functions on* $\mathcal{R}$, $\widehat{F}(\widehat{u}) = F(\widehat{u})$ in $L^\infty(0,T_{\mathrm{end}};\mathbf{H})$, hence in $L^2(0,T_{\mathrm{end}};\mathbf{V}')$; substituted into (S4) (an identity in $L^2(\mathbf{V}')$ tested against $v \in \mathbf{V}$, not pointwise), this makes $u := \widehat{u}$ a weak solution of (2.2) in the sense of Definition 2.2. The embedding $L^2(\mathbf{V}) \cap H^1(\mathbf{V}') \hookrightarrow C([0,T_{\mathrm{end}}];\mathbf{H})$ (Lions and Magenes 1972) gives meaning to $u(0) = u_0$ in $\mathbf{H}$.

Step 6 (uniqueness in the bounded class). Let $u, \bar{u}$ be bounded weak solutions (Definition 2.2) with the same datum. Lemma 3.2 (inputs: bounded class, $u_0 \in \mathcal{R}$) puts both in $\mathcal{R}$ a.e., so the Lipschitz estimate below holds on $\mathcal{R}$, not only along the constructed solution. Testing the difference of the weak formulations with $w = u - \bar{u}$:

$$\tfrac{1}{2}\tfrac{d}{dt}\|w\|^2_{\mathbf{H}} + d_*\|\nabla w\|^2_{\mathbf{H}} \;\le\; \big(F(u) - F(\bar{u}),\, w\big) \;\le\; L_{\mathcal{R}}\,\|w\|^2_{\mathbf{H}},$$

with $L_{\mathcal{R}}$ a Lipschitz constant of F on the compact convex set $\mathcal{R}$. By Cauchy–Schwarz, $(F(u) - F(\bar{u}),\, w) \le \|F(u) - F(\bar{u})\|_{\mathbf{H}}\, \|w\|_{\mathbf{H}}$, and $|F(u) - F(\bar{u})| \le \|DF\|_2\, |w|$ pointwise, so $L_{\mathcal{R}} = \sup_{\mathcal{R}} \|DF\|_2$ (spectral norm) $\le (\|DF\|_1 \|DF\|_\infty)^{1/2}$; the row and column sums of

$$DF = \begin{pmatrix} 1 - 2N - \delta H & 0 & -\delta N \\ -\alpha\beta T & \beta(1 - 2T - \alpha N) & 0 \\ 0 & \gamma & -\omega \end{pmatrix}$$

over $\mathcal{R}$ give

$$\begin{aligned} L_{\mathcal{R}} &= \big[\, \|DF\|_\infty\, \|DF\|_1 \,\big]^{1/2}, \qquad \text{with} \\ \|DF\|_\infty &= \max\big(1 + \delta(1 + M_0),\ \beta(1 + 2\alpha),\ \gamma + \omega\big), \\ \|DF\|_1 &= \max\big(1 + \delta M_0 + \alpha\beta,\ \beta(1 + \alpha) + \gamma,\ \delta + \omega\big). \end{aligned}$$

Gronwall and $w(0) = 0 \Rightarrow w \equiv 0$ on $[0, T_{\text{end}}]$; T_{end} is arbitrary and the bounds of Lemma 3.2 are uniform in time, so the solution is global. □

S2.5 Proof of Lemma 3.5

Proof of Lemma 3.5. Fix $t \ge 1$; on $(t-1, t)$, u weakly solves the linear Neumann problem with source $g := F(u) \in L^\infty(\Omega \times (t-1, t))$ and datum $u(t-1) \in \mathbf{H}$, attained in $\mathbf{H}$ (Lions and Magenes 1972). The linear uniqueness of Step 5 of the proof of Theorem 3.4 (Section S2.4), which uses only the energy identity, applies verbatim on $(t-1, t)$: u is the mild solution,

$$u(t) \;=\; e^{D\Delta_N}\, u(t-1) \;+\; \int_{t-1}^{t} e^{(t-s)D\Delta_N}\, F(u(s))\, ds, \qquad t \ge 1,$$

with $\{e^{\tau D\Delta_N}\}_{\tau \ge 0}$ the semigroup of Step 5. As $u \in \mathcal{R}$ (Theorem 3.4), $\|F(u(s))\|_{\mathbf{H}} \le C_F^\infty := \sup_{\mathcal{R}} |F| \cdot |\Omega|^{1/2}$, depending only on the parameters, $|\Omega|$ and M_0, not on t; likewise $\|u(t-1)\|_{\mathbf{H}} \le C_{\mathcal{R}}^\infty |\Omega|^{1/2}$, $C_{\mathcal{R}}^\infty := \sup_{v \in \mathcal{R}} |v| = (2 + M_0^2)^{1/2}$. The analytic-semigroup smoothing estimate (Henry 1981, Thm. 1.4.3),

$$\big\| e^{\tau D\Delta_N} \varphi \big\|_{[H^1(\Omega)]^3} \;\le\; c\,\big(1 + (d_*\tau)^{-1/2}\big)\, \|\varphi\|_{\mathbf{H}}, \qquad \tau > 0,$$

with $\tau = 1$ on the first term and $\tau = t - s \in (0, 1]$ under the integral, gives

$$\|u(t)\|_{[H^1(\Omega)]^3} \;\le\; c\,(1 + d_*^{-1/2})\, C_{\mathcal{R}}^\infty |\Omega|^{1/2} \;+\; c\, C_F^\infty \int_0^1 \big(1 + (d_*\tau)^{-1/2}\big) d\tau,$$

the integral being finite ($\tau^{-1/2}$ integrable at 0). This is (3.2), with a right-hand side K independent of $t \ge 1$ and depending only on α, β, γ, δ, ω, δ_1, δ_2, Ω and M_0; it depends on the diffusivities through $d_*^{-1/2}$, $d_* = \min(\delta_1, \delta_2, 1)$, which is large for the small diffusivities of the simulations (the dependence on M_0, hence on u_0, is removable by Remark 3.6). By Rellich–Kondrachov (Ω bounded, Lipschitz), $H^1(\Omega) \hookrightarrow L^2(\Omega)$ is compact, so the bounded set $\{u(t) : t \ge 1\} \subset [H^1(\Omega)]^3$ is precompact in $\mathbf{H}$. □

S3 Proof of the classification of equilibria

S3.1 Proof of Proposition 3.7

Proof of Proposition 3.7. From $\beta T(1 - T - \alpha N) = 0$, $T = 0$ or $T = 1 - \alpha N$; $\gamma T - \omega H = 0 \Rightarrow H = \gamma T/\omega$. If $T = 0$, then $H = 0$ and $N(1 - N) = 0$: E_0, E_1. If $T = 1 - \alpha N \ne 0$, substituting $H = \gamma(1 - \alpha N)/\omega$ into $N(1 - N - \delta H) = 0$ gives $N = 0$ (E_2) or $1 - N - (\delta\gamma/\omega)(1 - \alpha N) = 0$, i.e. the components (3.7).

For admissibility, write $p := \omega - \delta\gamma$ and $q := \omega - \alpha\delta\gamma$, so that

$$N^* = \frac{p}{q}, \quad T^* = \frac{\omega(1 - \alpha)}{q}, \quad H^* = \frac{\gamma(1 - \alpha)}{q}, \qquad q - p = \delta\gamma\,(1 - \alpha). \tag{S5}$$

Now $T^*, H^* > 0 \iff (1-\alpha)/q > 0 \iff \operatorname{sign} q = \operatorname{sign}(1-\alpha)$; $N^* > 0 \iff \operatorname{sign} p = \operatorname{sign} q$. Hence all three are positive iff $\operatorname{sign} p = \operatorname{sign} q = \operatorname{sign}(1-\alpha)$, and by (S5), $\operatorname{sign} p = \operatorname{sign}(1-\alpha) \neq 0$ already implies the same sign for q (the other pairs do not imply the third condition). If $\alpha < 1$, this reduces to $p > 0$, i.e. $\delta < \delta_c$, and then $q = p + \delta\gamma(1-\alpha) > 0$. If $\alpha > 1$, it reduces to $p < 0$, i.e. $\delta > \delta_c$, and then $q = p + \delta\gamma(1-\alpha) < 0$, both summands being negative. The case $\alpha = 1$ forces $T^* = H^* = 0$, $N^* = 1$ (E_3 collapses onto E_1); $\delta = \delta_c$ forces $N^* = 0$ (E_3 collapses onto E_2). This proves (3.8).

Finally, $E_3 \in \mathcal{R}$ on both branches, since $|p| < |q|$ by (S5): $0 < p < q$ on branch (i), $q < p < 0$ on branch (ii). Hence $N^* = p/q \in (0,1)$, so $T^* = 1 - \alpha N^* < 1$ and $H^* = \gamma T^*/\omega \in (0, \gamma/\omega)$, both strictly. □

S4 Coefficient expansion for spatial stability

The degree-by-degree expansion below is what the closed form of Proposition 3.10 replaces.

Let E_3 lie on branch (i), write $J_k = J(E_3) - \mu D$ with $D = \operatorname{diag}(\delta_1, \delta_2, 1)$ and $\mu = \mu_k \geq 0$ an eigenvalue of $-\Delta_N$, and set

$$A = N^* + \delta_1\mu, \qquad B = \beta T^* + \delta_2\mu, \qquad C = \omega + \mu, \qquad \kappa_3 = \alpha\beta\delta\gamma\, N^* T^* > 0.$$

The characteristic polynomial of J_k is $\lambda^3 + \tilde{a}_1\lambda^2 + \tilde{a}_2\lambda + \tilde{a}_3$ with $\tilde{a}_1 = A+B+C$, $\tilde{a}_2 = AB+BC+CA$ and $\tilde{a}_3 = ABC - \kappa_3$, and the middle coefficient receives no cross term precisely because the interaction graph has no 2-cycle. Expanding in powers of μ gives

$$\tilde{a}_1(\mu) = a_1 + \mu(\delta_1 + \delta_2 + 1), \tag{S6}$$

$$\tilde{a}_2(\mu) = a_2 + \mu\big[N^*(\delta_2+1) + \beta T^*(\delta_1+1) + \omega(\delta_1+\delta_2)\big] + \mu^2(\delta_1\delta_2 + \delta_1 + \delta_2), \tag{S7}$$

$$\tilde{a}_3(\mu) = a_3 + \mu\big[N^*\beta T^* + \omega(N^*\delta_2 + \beta T^*\delta_1)\big] + \mu^2\big[N^*\delta_2 + \beta T^*\delta_1 + \delta_1\delta_2\omega\big] + \mu^3\delta_1\delta_2, \tag{S8}$$

and each difference $\tilde{a}_i(\mu) - a_i$ is a polynomial with strictly positive coefficients. The closed form is shorter: A, B, C increase with μ, and the identity $(A+B+C)(AB+BC+CA) - ABC = (A+B)(B+C)(C+A)$ settles the second Routh–Hurwitz condition (both checked in `sympy_verification.py`, Section 9).

Remark S4.1 (Why the absence of a two-cycle is what matters). Two structural facts produce (3.16). The diagonality of D confines $-\mu D$ to the diagonal of J_3, so the whole μ-dependence sits in the increasing A, B, C of (3.14). The absence of a 2-cycle (each coupling acts one way only) leaves $\tilde{a}_2 = AB + BC + CA$ without cross-correction, hence increasing in μ, which diagonality alone would not ensure. The single 3-cycle $N \to T \to H \to N$ contributes the constant $\kappa_3 = \alpha\beta\delta\gamma N^*T^*$, subtractively to $\tilde{a}_3 = ABC - \kappa_3$ and additively to $\tilde{a}_1\tilde{a}_2 - \tilde{a}_3 = (A+B)(B+C)(C+A) + \kappa_3$: every Routh–Hurwitz quantity is non-decreasing in μ, whatever the diffusivities. This makes the coefficients, and μ^* of Proposition 3.11, explicit; the exclusion itself follows from Proposition 3.12, since subtracting $\mu D \geq 0$ from the Metzler matrix $\Sigma J \Sigma$ lowers its spectral bound. A symmetric term $-\alpha' NT$ in the normal-cell equation keeps $(\Sigma J_3 \Sigma)_{12} = \alpha' N \geq 0$; only a non-diagonal D or a coupling breaking cooperativity would reopen the question.

S4.1 Functional setting for the invariant manifolds

To speak of invariant manifolds of a constant steady state, and not only of signs of eigenvalues, we fix the setting. The operator $\mathcal{A} := D\Delta_N + J(E^*)$, with Δ_N the Neumann Laplacian on Ω, is a bounded perturbation of a diagonal sectorial operator; it is therefore sectorial on $\mathbf{H} = [L^2(\Omega)]^3$ and generates an analytic semigroup. Since the reaction field is smooth, the semiflow of (2.2) is C^1 near E^* in the fractional power space $\mathbf{X}^\theta = \mathcal{D}((-\mathcal{A}_0)^\theta)$, built on the self-adjoint part $\mathcal{A}_0 := D\Delta_N - I$, with $\theta \in (\frac{3}{4}, 1)$ (Henry 1981, Ch. 1 and 3). The statements below use the embedding $\mathbf{X}^\theta \hookrightarrow [L^\infty(\Omega)]^3$, which requires more of Ω than the Lipschitz regularity assumed in Section 2.3; we therefore state the extra hypothesis explicitly. It follows from $\mathbf{X}^\theta \hookrightarrow [H^{2\theta}(\Omega)]^3$ together with $2\theta > d/2$, and the first inclusion needs $\mathcal{D}(\Delta_N) = \{v \in H^2(\Omega) : \partial_\nu v = 0\}$, which holds when Ω is convex or of class $C^{1,1}$ (Grisvard 2011, §3.2) and fails on a general Lipschitz domain, where the operator domain lies only in $H^{3/2}$. We therefore assume Ω convex throughout what follows and in Remark 3.13; the square $(0, L)^2$ of

Section 4 satisfies this assumption, which Remark 3.13 also needs for a separate reason. Both D and Δ_N act diagonally on the eigenfunctions of $-\Delta_N$, so $\mathcal{A}$ is the orthogonal sum $\bigoplus_k (J(E^*) - \mu_k D)$ and $\sigma(\mathcal{A}) = \bigcup_k \sigma(J(E^*) - \mu_k D)$, a closed set: the k-th block satisfies $\operatorname{Re}\lambda \le -\min(\delta_1, \delta_2, 1)\,\mu_k + \|J(E^*)\|$, so only finitely many blocks meet any bounded region. When $\sigma(\mathcal{A})$ misses the imaginary axis, the local stable and unstable manifold theorem for sectorial operators (Henry 1981, §5.1–5.2) applies, and $W^u(E^*)$ is finite-dimensional; its dimension is the number of eigenvalues with positive real part, counted with algebraic multiplicity, and all statements below about $\dim W^u$ refer to this number.

S4.2 Strong monotonicity and order relations

This is the positivity step of the proof of Proposition 3.12. Two ordered solutions differ by $e = \Sigma(v-u)$, which solves a linear cooperative system; the lower bound of the Neumann heat kernel (Gyrya and Saloff-Coste 2011) makes its non-trivial components positive on $\bar\Omega$ for $t > 0$, and the coupling, irreducible where $N, T > 0$, does the rest, with no boundary point lemma. The relation $x \le_K y$, that is $\Sigma(y-x) \ge 0$ a.e., is read in $\mathbf{H}$; the strict relation $x \ll_K y$ is read in $\mathbb{R}^3$ or in $\mathbf{X}^\theta \hookrightarrow [C(\bar\Omega)]^3$, K having empty interior in $\mathbf{H}$.

S5 Extended codim-2 unfolding and degenerate cases

S5.1 Proof of Proposition 3.19

Proof of Proposition 3.19. Substituting $N = s$, $T = 1-s$, $H = \gamma(1-s)/\omega$, $\alpha = 1$ and $\delta = \delta_c = \omega/\gamma$ into (2.3) gives $\gamma T - \omega H = 0$, $\beta(1-s)\big(1-(1-s)-s\big) = 0$, $s\big(1-s-(1-s)\big) = 0$; and $\mathcal{S} \subset \mathcal{R}$ ($s \in [0,1]$, $\gamma(1-s)/\omega \le \gamma/\omega \le M_0$). Conversely, $F = 0$ gives $T \in \{0, 1-N\}$ and $H = \gamma T/\omega$; $T = 0$ $\Rightarrow H = 0$, $N(1-N) = 0$ (E_0, E_1); $T = 1-N \Rightarrow N\big(1-N-(1-N)\big) = 0$ identically. The zero set is thus $\Lambda \cup \{E_0\}$, where

$$\Lambda := \Big\{\big(s,\ 1-s,\ \tfrac{\gamma}{\omega}(1-s)\big)\ :\ s \in \mathbb{R}\Big\} \tag{S9}$$

is the affine line carrying $\mathcal{S}$; the zero set $\Lambda \cup \{E_0\}$ meets $\mathcal{R}$ in $\mathcal{S} \cup \{E_0\}$. At the point s of $\mathcal{S}$, the Jacobian (3.9) is

$$J(s) = \begin{pmatrix} -s & 0 & -\omega s/\gamma \\ -\beta(1-s) & -\beta(1-s) & 0 \\ 0 & \gamma & -\omega \end{pmatrix}, \tag{S10}$$

and the tangent $\tau = (1,\ -1,\ -\gamma/\omega)^\top$ of $\mathcal{S}$ satisfies $J(s)\tau = 0$ (row by row): 0 is an eigenvalue, with eigenvector τ. The characteristic polynomial of (S10) is $\lambda\big(\lambda^2 + A(s)\lambda + B(s)\big)$ with

$$A(s) = s + \beta(1-s) + \omega, \qquad B(s) = \omega s + \beta(1-s)(\omega + s). \tag{S11}$$

On $[0,1]$, $A > 0$ (terms ≥ 0, $\omega > 0$) and $B > 0$ (its first term vanishes only at $s = 0$, where the second is $\beta\omega > 0$; the second only at $s = 1$, where the first is $\omega > 0$). Hence the other two eigenvalues have negative real part, 0 is simple, and $\mathcal{S}$ is normally hyperbolic and attracting. □

S5.2 Degenerate cases

On the boundary of the existence region of E_3 the formulas (3.7) produce coalescing equilibria rather than new branches. At $\alpha = 1$ (the transcritical curve Γ_α) they give $E_3 = E_1$, reached from branch (i) when $\delta < \delta_c$ and from branch (ii) otherwise; at $\delta = \delta_c$ (the curve Γ_δ) they give $E_3 = E_2$, reached according to the sign of $1-\alpha$. At $P^* = (1, \delta_c)$, where the curves meet and the coexistence state changes type, neither collapse can be read off the formulas, because the zero set contains the whole segment $\mathcal{S}$ of Proposition 3.19. Finally, the denominator $q = \omega - \alpha\delta\gamma$ vanishes on the hyperbola $\delta = \delta_c/\alpha$, which lies strictly above δ_c when $\alpha < 1$ and strictly below it when $\alpha > 1$, that is, where E_3 is inadmissible: the singularity is only apparent.

Sotomayor coefficients on Γ_α. At $E_1 = (1,0,0)$ for $\alpha = 1$, with α as the bifurcation parameter, the right and left null vectors of $J(E_1; \alpha = 1)$ are $v = (-\delta, \omega/\gamma, 1)^\top$ and $w = (0,1,0)^\top$, and

$$w^\top F_\alpha = 0, \qquad w^\top (DF)_\alpha v = -\beta\omega/\gamma \ne 0, \qquad w^\top D^2F(v,v) = 2\beta\omega(\delta\gamma - \omega)/\gamma^2.$$

For $\delta \neq \delta_c$ the bifurcation is a non-degenerate transcritical exchange of stability between E_1 and E_3; at $\delta = \delta_c$ the quadratic coefficient vanishes, which flags P^*. Its sign has no invariant meaning, since v and w are defined only up to scalars; the convention-free quantity is the slope $-2\sigma_1 v/\sigma_2$ of the bifurcating branch, which on Γ_δ returns $\mathrm{d}N^*/\mathrm{d}\delta = -\gamma/(\omega(1-\alpha))$, in agreement with the closed form (script, Section 8).

S5.3 Unfolding of the codimension-two point

These are the computations summarised in Section 3.6. Writing $\alpha = 1 + a$ and $\delta = \delta_c(1 + d)$, the components (3.7) become

$$N^* = \frac{d}{a + d + ad}, \qquad T^* = \frac{a}{a + d + ad}, \qquad H^* = \frac{\gamma}{\omega}\,\frac{a}{a + d + ad}, \tag{S12}$$

whose common denominator is $-q/\omega$. It vanishes on a curve with tangent $d = -a$, along which E_3 is unbounded. Along a ray $(a, d) = \varepsilon(a_0, d_0)$ with $a_0 + d_0 \neq 0$ one has $N^* + T^* \to 1$, and

$$\lim_{\varepsilon \downarrow 0} E_3 = \Big(s_0,\ 1 - s_0,\ \tfrac{\gamma}{\omega}(1 - s_0)\Big), \qquad s_0 = \frac{d_0}{a_0 + d_0}, \tag{S13}$$

which is a point of $\mathcal{S}$ exactly when $a_0 d_0 \geq 0$, that is, on the closure of the admissible cone; as the ray turns, s_0 sweeps $[0, 1]$, so E_3 spreads over $\mathcal{S}$ instead of converging to a point.

The centre direction at P^* is $\mathcal{S}$ itself, so the unfolding needs no normal form, but persistence needs care: the theorems of Fenichel and of Hirsch, Pugh and Shub (1971; 1977), see also Wiggins (1994), require a compact manifold *with boundary* to be crossed transversally at its boundary, whereas the field vanishes at E_1 and E_2. Nothing in the proof of Proposition 3.19 used $s \in [0, 1]$: the field vanishes on the whole line (S9), and $A(s)$, $B(s)$ stay positive on some $[-\varepsilon_0, 1 + \varepsilon_0]$. On the enlarged segment, the transverse eigenvalues satisfy $\operatorname{Re}\lambda \leq -c$, and the field may be modified in the buffer, away from $\Lambda|_{[0,1]}$, to point strictly outward; the modified system has an overflowing invariant normally hyperbolic manifold, and the conclusion over $[0, 1]$ does not depend on the modification. For (a, d) near the origin there is therefore a locally invariant attracting curve $\mathcal{S}_{a,d}$, of class C^r for each finite r and $O(|a| + |d|)$-close to $\mathcal{S}$, which has endpoints E_1 and E_2 and is the maximal invariant set in its neighbourhood.

On $\mathcal{S}$ the perturbed field is

$$F\big(s,\ 1 - s,\ \tfrac{\gamma}{\omega}(1 - s);\ 1 + a,\ \delta_c(1 + d)\big) = -\,s(1 - s)\,\big(d,\ a\beta,\ 0\big)^{\top}, \tag{S14}$$

and projecting along the stable fibres with the left null vector $\ell(s) = \big(-\beta(1 - s),\ s,\ \beta s(1 - s)/\gamma\big)$ of $J(s)$ gives the reduced field (3.22). The remainder is $O(\varepsilon^2)$, where $\varepsilon = |a| + |d|$, because the first-order correction to the parameterisation $x_{a,d}(s) = x(s) + \varepsilon h(s) + O(\varepsilon^2)$ carries the factor $\ell(s)J(s) = 0$. The reduced field vanishes at $s = 0$ and $s = 1$ (E_2 and E_1) and at $s^* = d/(a + d)$, which equals N^* up to $O(\varepsilon)$ and lies in $(0, 1)$ exactly when a and d have the same sign. Since the bracket is affine in s with slope $a + d$, s^* attracts when $a + d < 0$ and repels when $a + d > 0$. The linearisation there is $O(\varepsilon)$ and the remainder is $O(\varepsilon^2)$ in C^1, because Fenichel's theorem yields the reduced field as a C^r object depending C^r on ε; the sign therefore survives: branch (i) carries a stable coexistence state and branch (ii) a saddle between two stable endpoints. At the endpoints, $\kappa(0) = \beta$ and $\kappa(1) = 1$ give $\dot{s} = -d\,s$ and $\dot{\sigma} = -\beta a\,\sigma$ ($\sigma = 1 - s$), which are exactly the kinetic eigenvalues of E_2 and E_1, as they must be, since both points lie on $\mathcal{S}_{a,d}$.

Section 11 of `sympy_verification.py` checks these identities and runs a numerical control. The full kinetics and the reduced equation are integrated from $s = \frac{1}{2}$ at $(a, d) = \varepsilon(a_0, d_0)$, for $\varepsilon \in \{10^{-2}, 5 \cdot 10^{-2}\}$ and $(a_0, d_0) \in \{(1, 2), (-1, -2), (1, \frac{1}{2})\}$; the two select the same equilibrium in all six cases, and the residual gap on branch (i) is exactly $s^* - N^* = ad^2/[(a + d)(a + d + ad)]$ ($-2.30 \cdot 10^{-2}$ at $\varepsilon = 5 \cdot 10^{-2}$).

S6 Proofs of the global-stability results

S6.1 Proof of Lemma 3.14

Proof of Lemma 3.14. A face-by-face argument misses the edges, so we bound the six distances to the boundary from below along the orbit. Fix $u_0 \in \mathcal{B}_\eta$ and set

$$\tau := \sup\{t > 0 : u(s) \in \mathcal{B}_\eta \text{ for every } s \in [0,t)\},$$

$\tau > 0$ since $\mathcal{B}_\eta$ is open and u continuous. All inequalities below hold on $[0,\tau)$, where $u \in \mathcal{B}_\eta$. There $1 - N - \delta H \geq -c$, $c := 1 + \delta(\gamma/\omega + \eta)$, so $\dot{N} \geq -cN$ and $N(t) \geq N(0)e^{-ct}$; likewise $\dot{T} \geq -\beta c' T$, $c' := 1 + \alpha$, $\Rightarrow T(t) \geq T(0)e^{-\beta c' t}$, and $\dot{H} \geq -\omega H \Rightarrow H(t) \geq H(0)e^{-\omega t}$. For the upper bounds, $w := 1 - N$, $v := 1 - T$, $z := \gamma/\omega + \eta - H$ satisfy on $[0,\tau)$

$$\dot{w} = -Nw + \delta NH \geq -w, \qquad \dot{v} = -\beta T v + \alpha\beta TN \geq -\beta v,$$

$$\dot{z} = \gamma(1 - T) + \omega\eta - \omega z \geq -\omega z,$$

since $N, T \leq 1$ and the dropped terms δNH, $\alpha\beta TN$, $\gamma(1-T) + \omega\eta$ are ≥ 0 on $\mathcal{B}_\eta$; by Gronwall, $w(t) \geq w(0)e^{-t}$, $v(t) \geq v(0)e^{-\beta t}$, $z(t) \geq z(0)e^{-\omega t}$. Suppose $\tau < \infty$. The orbit is bounded on $[0,\tau)$, so by the blow-up alternative $u(\tau)$ is defined, and by definition of τ, $u(\tau) \in \partial\mathcal{B}_\eta$: one of N, T, H, w, v, z vanishes at τ. But each is $\geq$ its (positive) value at $t = 0$ times one of $e^{-c\tau}$, $e^{-\beta c'\tau}$, $e^{-\omega\tau}$, $e^{-\tau}$, $e^{-\beta\tau}$, a contradiction; hence $\tau = +\infty$. E_0, E_1, E_2 have a vanishing or saturated component, so lie outside $\mathcal{B}_\eta$; $E_3 \in \mathcal{B}_\eta$ by the strict inequalities of the proof of Proposition 3.12 and $H^* < \gamma/\omega \leq \gamma/\omega + \eta$. At P^* the statement fails: the whole segment of Proposition 3.19 lies in $\mathcal{B}_\eta$. □

S6.2 Proof of Theorem 3.15

Proof of Theorem 3.15. We squeeze u between two explicit constant states moved monotonically by the flow. The equilibria are those of Proposition 3.7: the three cases avoid Γ_α and Γ_δ by their defining inequalities, and on $\{q = 0\}$, which meets cases *(ii)*, *(iii)* but neither admissibility branch, E_3 is undefined and only E_0, E_1, E_2 remain. Only monotonicity of the semiflow is used (neither strong monotonicity nor the order interval): $F(\bar{x}) \in -K \Rightarrow t \mapsto \varphi_t(\bar{x})$ is K-decreasing, $F(\underline{x}) \in K \Rightarrow$ K-increasing (Smith 1995, Prop. 3.2.1). It applies for two reasons. First, $\mathcal{Q}_\eta := [0,1]^2 \times [0, \gamma/\omega + \eta]$ is positively invariant for the kinetics: the sign checks of Theorem 3.1 hold verbatim on $\{N = 0\}$, $\{N = 1\}$, $\{T = 0\}$, $\{T = 1\}$, $\{H = 0\}$, and on $\{H = \gamma/\omega + \eta\}$, $F_3 = \gamma T - \gamma - \omega\eta \leq -\omega\eta \leq 0$. The constant states below lie in $\mathcal{Q}_\eta$, so their orbits are bounded, and a bounded orbit with monotone components converges to a zero of F. Second, the quasimonotonicity of Proposition 3.12 holds on $\{N \geq 0,\ T \geq 0\}$ only (two of the three non-zero off-diagonal entries of $\Sigma J \Sigma$ are δN, $\alpha\beta T$); $\mathcal{Q}_\eta$ is convex and contained in that half-space, so all compared states and the segments joining them stay where the principle holds.

The upper barrier, for $\alpha < 1$. Let $\bar{u}_\varepsilon := \big(1 - \delta\gamma\varepsilon/\omega,\ \varepsilon,\ \gamma\varepsilon/\omega\big)$; then $F_1 = F_3 = 0$ and

$$F(\bar{u}_\varepsilon) = \Big(0,\ \beta\varepsilon\big(1 - \alpha - \varepsilon + \tfrac{\alpha\delta\gamma}{\omega}\varepsilon\big),\ 0\Big) \in -K \tag{S15}$$

for ε small enough that the bracket is ≥ 0, possible exactly because $\alpha < 1$. So the orbit through $\bar{u}_\varepsilon$ is K-decreasing ($N \downarrow$, $T, H \uparrow$), bounded, and converges to an equilibrium with $T \geq \varepsilon > 0$: E_2 or E_3. In case *(ii)* E_3 is inadmissible, so the limit is E_2. In case *(i)* it is not E_2: $N \downarrow 0$, $H \uparrow \gamma/\omega$ would give $1 - N - \delta H \to 1 - \delta/\delta_c > 0$, so $\dot{N} > 0$ for large t, contradicting $N \downarrow$. The limit is E_3.

The lower barrier, for $\delta < \delta_c$. Let $\underline{u}_\varepsilon := \big(\varepsilon,\ 1 - \alpha\varepsilon,\ \gamma/\omega + \eta\big)$; then $F_2 = 0$ and

$$F(\underline{u}_\varepsilon) = \Big(\varepsilon\big(1 - \varepsilon - \delta(\tfrac{\gamma}{\omega} + \eta)\big),\ 0,\ -\alpha\gamma\varepsilon - \omega\eta\Big) \in K \tag{S16}$$

for $\varepsilon < 1 - \delta(\gamma/\omega + \eta)$, a non-empty condition since $\eta < \eta_0$ and $\delta < \delta_c$. This barrier sits on $\partial\mathcal{B}_\eta$ ($H = \gamma/\omega + \eta$), but $\dot{H} = -\alpha\gamma\varepsilon - \omega\eta < 0$ there, so the orbit enters $\mathcal{B}_\eta$ at once and Lemma 3.14 applies for $t > 0$. The orbit is K-increasing with $N \uparrow$ from ε, so its limit has $N > 0$: E_1 or E_3. In

case *(iii)* E_3 is inadmissible and the limit is E_1; in case *(i)* it is not E_1, since $T \downarrow 0$, $N \uparrow 1$ would give $\dot{T}/T \to \beta(1-\alpha) > 0$, contradicting $T \downarrow$; so the limit is E_3.

The squeeze. As $\varepsilon \downarrow 0$ the barriers tend to $E_1 = (1,0,0)$ and $(0,1,\gamma/\omega+\eta)$, the supremum and infimum of $\mathcal{B}_\eta$ for $\leq_K$: for $u_0 = (N_0, T_0, H_0) \in \mathcal{B}_\eta$ and ε small, the six inequalities $\varepsilon < N_0 < 1 - \delta\gamma\varepsilon/\omega$, $\varepsilon < T_0 < 1 - \alpha\varepsilon$ and $\gamma\varepsilon/\omega < H_0 < \gamma/\omega + \eta$ read $\underline{u}_\varepsilon \ll_K u_0 \ll_K \bar{u}_\varepsilon$. In case *(i)* both barriers exist and monotonicity gives $\varphi_t(\underline{u}_\varepsilon) \leq_K u(t) \leq_K \varphi_t(\bar{u}_\varepsilon)$, $t \geq 0$, both bounds $\to E_3$. In case *(ii)* the lower barrier is replaced by the explicit solution from $(0,1,\gamma/\omega+\eta)$, $N \equiv 0$, $T \equiv 1$, $H(t) = \gamma/\omega + \eta e^{-\omega t}$, K-below u_0 and $\to E_2$; with the upper barrier, $u(t) \to E_2$. In case *(iii)* the upper barrier is replaced by the equilibrium E_1, K-above u_0; with the lower barrier, $u(t) \to E_1$. □

S6.3 Proof of Theorem 3.16

Proof of Theorem 3.16. All quantities below are essential infima or suprema of the bounded measurable N, T, H; no pointwise regularity is needed.

Placing a constant pair around the solution. Fix $\eta \in (0, \eta_0)$ in cases *(i)*, *(iii)*, $\eta > 0$ arbitrary in case *(ii)*, and $t_1 > 0$ with $\|H(\cdot,t_1)\|_\infty < \gamma/\omega + \eta$ (Remark 3.6). Since this bound is approached from above when $\|H_0\|_\infty > \gamma/\omega$, $\mathcal{B}_\eta$, not $\mathcal{B}_0$, is the right receptacle. Positivity at t_1, on which all six bounds below rest, is where connectedness enters. Write $\partial_t N = \delta_1 \Delta N + c\,N$, $c := 1 - N - \delta H$, $|c| \leq \kappa := 1 + \delta M_0$ on $\mathcal{R}$. Then $e^{\kappa t} N \geq 0$ is a supersolution of the Neumann heat equation, so $N(\cdot,t) \geq e^{-\kappa t}\, e^{t\delta_1 \Delta_N} N_0$ a.e. A bounded connected Lipschitz domain is inner uniform, so the Neumann heat kernel has two-sided Gaussian bounds, hence a lower bound $p_t > 0$ on $\Omega \times \Omega$, $\forall t > 0$ (Gyrya and Saloff-Coste 2011), whence

$$\operatorname*{ess\,inf}_\Omega N(\cdot,t_1) \;\geq\; e^{-\kappa t_1} p_{t_1} \int_\Omega N_0 \;>\; 0,$$

as $N_0 \geq 0$, $N_0 \not\equiv 0$. Likewise $\operatorname{ess\,inf}_\Omega T(\cdot,t_1) > 0$, with $c := \beta(1 - T - \alpha N)$.

The other three quantities may vanish identically at $t = 0$, so we use Duhamel instead. The acid solves $\partial_t H = \Delta H - \omega H + \gamma T$, $H_0 \geq 0$ (possibly $\equiv 0$, as in all our simulations), so

$$H(\cdot,t_1) \;\geq\; \gamma \int_0^{t_1} e^{-\omega(t_1-s)} e^{(t_1-s)\Delta_N} T(\cdot,s)\,\mathrm{d}s,$$

bounded away from zero since $T(\cdot,s)$ is, uniformly for $s \in [t_1/2, t_1]$: the lower bound obtained for T at t_1 holds with the same constant on that compact time interval, the heat-kernel bound being uniform there. Same for the upper bounds: $w := 1 - N$, $v := 1 - T$, $z := \gamma/\omega + \eta - H$ are ≥ 0 at t_1 by choice of t_1 and invariance of $\mathcal{R}$; w, v may vanish identically at $t = 0$ (tissue at carrying capacity), and

$$\partial_t w - \delta_1 \Delta w + N w = \delta N H, \qquad \partial_t v - \delta_2 \Delta v + \beta T v = \alpha\beta T N, \qquad \partial_t z - \Delta z + \omega z = \gamma v + \omega\eta,$$

with right-hand sides ≥ 0, and bounded away from zero for w, v once N, T, H are. Duhamel, as for H, gives $\operatorname{ess\,inf}_\Omega w(\cdot,t_1) > 0$ and $\operatorname{ess\,inf}_\Omega v(\cdot,t_1) > 0$; and $z(\cdot,t_1) \geq \gamma/\omega + \eta - \|H(\cdot,t_1)\|_\infty > 0$ by definition of t_1. Hence

$$m := \big(\operatorname{ess\,inf} N,\ \operatorname{ess\,sup} T,\ \operatorname{ess\,sup} H\big)\Big|_{t=t_1}, \qquad M := \big(\operatorname{ess\,sup} N,\ \operatorname{ess\,inf} T,\ \operatorname{ess\,inf} H\big)\Big|_{t=t_1}$$

both lie in $\mathcal{B}_\eta$, and $m \leq_K u(\cdot,t_1) \leq_K M$ a.e.

Propagating the enclosure. Spatially constant solutions of the kinetics solve (2.2) (Δ annihilates them, Neumann condition holds), so $t \mapsto \varphi_{t-t_1}(m)$, $t \mapsto \varphi_{t-t_1}(M)$ (φ the kinetic flow of Theorem 3.15) solve the full system and stay in $\mathcal{B}_\eta$ by Lemma 3.14. We compare them with u in the weak form of Theorem 3.4. With $\Sigma = \operatorname{diag}(1,-1,-1)$ (Proposition 3.12), $x \leq_K y \iff \Sigma(y-x) \geq 0$ componentwise; set

$$e \;:=\; \Sigma\big(\varphi_{t-t_1}(m) - u\big).$$

Subtracting the equations, with $F(a) - F(b) = A(a,b)\,(a-b)$, $A(a,b) = \int_0^1 DF\big(b + s(a-b)\big)\,\mathrm{d}s$ (licit: $\mathcal{Q}_\eta$ is convex and contains both arguments, $u(\cdot,t) \in \mathcal{Q}_\eta$ for $t \geq t_1$ because $\gamma/\omega + \eta$ is a supersolution of the H-equation when $T \leq 1$), gives

$$\partial_t e \;=\; D\Delta e \;+\; B\,e, \qquad B := \Sigma A \Sigma,$$

as the diagonal D, Σ commute. By Proposition 3.12, $\Sigma\, DF\, \Sigma$ has off-diagonal entries ≥ 0 on $\{N, T \geq 0\} \supset \mathcal{Q}_\eta$, hence so has B, an average of such matrices. Testing against e^+, with the chain rule for $\frac{\mathrm{d}}{\mathrm{d}t}\|e^+\|_{L^2}^2$ of Alt and Luckhaus (1983, Lemma 1.5) and no boundary term (Neumann),

$$\tfrac{1}{2}\tfrac{\mathrm{d}}{\mathrm{d}t}\|e^+\|_{L^2}^2 \;=\; -\int_\Omega D\,|\nabla e^+|^2 \;+\; \int_\Omega e^+ \cdot Be \;\leq\; C\,\|e^+\|_{L^2}^2,$$

since the first term is ≤ 0, $e_i^+ e_i = (e_i^+)^2$, $B_{ij} e_j \leq B_{ij} e_j^+$ for $i \neq j$ ($B_{ij} \geq 0$), and $C := \sup_{\mathcal{Q}_\eta} |DF| < \infty$ (F smooth on a bounded set). As $e^+(\cdot, t_1) = 0$, Gronwall gives $e^+ \equiv 0$, i.e. $\varphi_{t-t_1}(m) \leq_K u(\cdot, t)$; the same computation on $\Sigma\big(u - \varphi_{t-t_1}(M)\big)$ gives the upper enclosure. Hence $\varphi_{t-t_1}(m) \leq_K u(\cdot,t) \leq_K \varphi_{t-t_1}(M)$ for $t \geq t_1$, a.e. on Ω, and by Theorem 3.15 on $\mathcal{B}_\eta$ both bounds converge to the same equilibrium E. Each component of u is thus squeezed between two spatially constant functions with a common limit: $\|u(\cdot,t) - E\|_{L^\infty(\Omega)} \to 0$. □

S6.4 Proof of Theorem 3.17

Proof of Theorem 3.17. The bootstrap "$T \to 1 \Rightarrow H \to \gamma/\omega \Rightarrow N \to 0$" is circular, its first step presupposing $N \to 0$. A weaker, unconditional lower bound on T breaks the circle and still drains N. All comparisons use spatially homogeneous sub-/supersolutions, i.e. solutions of scalar ODEs on a Neumann domain.

Step 1 (unconditional lower bound $\liminf T \geq 1 - \alpha$*).* As $0 \leq N \leq 1$ (Lemma 3.2),

$$\partial_t T \;\geq\; \beta\, T\,(1 - \alpha - T) + \delta_2 \nabla^2 T.$$

The solution $\underline{v}$ of $\underline{v}' = \beta\,\underline{v}\,(1 - \alpha - \underline{v})$, $\underline{v}(0) = \min(c_0, 1-\alpha) > 0$, is a homogeneous subsolution with $\underline{v}(0) \leq T_0$, so $T(x,t) \geq \underline{v}(t)$ a.e. (Lemma 3.3), and $\underline{v}(t) \to 1 - \alpha$, stable since $\alpha < 1$. Thus $\forall \varepsilon > 0$ $\exists\, t_1(\varepsilon)$:

$$T(x,t) \;\geq\; 1 - \alpha - \varepsilon \qquad \text{for all } x \in \Omega,\ t \geq t_1. \tag{S17}$$

Only this step uses $T_0 \geq c_0 > 0$, to keep $\underline{v}(0)$ off the unstable equilibrium 0.

Step 2 (lower bound on the acid). Let $\underline{w}(t) = \gamma(1 - \alpha - \varepsilon)/\omega$ (constant). By (S17), for $t \geq t_1$, $\partial_t H - \nabla^2 H = \gamma T - \omega H \geq \gamma(1 - \alpha - \varepsilon) - \omega H$: H is a supersolution of the linear problem relaxing to $\underline{w}$. Comparison (Lemma 3.3) with the solution of $y' = \gamma(1 - \alpha - \varepsilon) - \omega y$, $y(t_1) = 0$, which tends to $\underline{w}$ at rate $e^{-\omega(t - t_1)}$, yields

$$H(x,t) \;\geq\; \frac{\gamma(1 - \alpha - \varepsilon)}{\omega} \;-\; C\, e^{-\omega(t - t_1)}, \qquad t \geq t_1, \tag{S18}$$

with $C = \gamma(1 - \alpha - \varepsilon)/\omega \leq \gamma(1 - \alpha)/\omega$; so $\liminf_t \inf_x H \geq \gamma(1 - \alpha)/\omega$ as $\varepsilon \downarrow 0$.

Step 3 (the normal tissue drains, exponentially and in L^∞*).* Insert (S18) into the N-equation, drop $-N^2 \leq 0$, and take $t \geq t_2 \geq t_1$ with the exponential in (S18) below ε:

$$\partial_t N \;\leq\; N\Big(1 - \frac{\delta\gamma(1 - \alpha - \varepsilon)}{\omega} + \delta\varepsilon\Big) + \delta_1 \nabla^2 N \;=\; -\mu\, N + \delta_1 \nabla^2 N,$$

where ($\delta_c = \omega/\gamma$)

$$\mu \;:=\; \frac{\delta\,(1 - \alpha - \varepsilon)}{\delta_c} \;-\; 1 \;-\; \delta\varepsilon.$$

The hypothesis $\delta(1 - \alpha) > \delta_c$, used only here, is exactly $\mu > 0$ for small ε (negative effective growth rate of N), and $\mu \uparrow \mu_0 = \delta(1 - \alpha)/\delta_c - 1$ as $\varepsilon \downarrow 0$, so every $\mu \in (0, \mu_0)$ is reached. The times t_1 and t_2 depend only on c_0, ε and the parameters: t_1 through the logistic subsolution started at $\min(c_0, 1 - \alpha)$, and t_2 through the explicit exponential in (S18). $\overline{z}(t) = e^{-\mu(t - t_2)}$ solves $\overline{z}' = -\mu\overline{z}$, $\overline{z}(t_2) = 1 \geq N(t_2)$, so it is a homogeneous supersolution and Lemma 3.3 gives

$$\|N(t)\|_{L^\infty(\Omega)} \;\leq\; e^{-\mu\,(t - t_2)} \qquad (t \geq t_2). \tag{S19}$$

The decay holds directly in L^∞, with explicit rate and no parabolic smoothing.

Step 4 (closing the loop). By (S19), $\alpha N(t) \to 0$ uniformly in x, so

$$\beta\, T\,(1-\vartheta(t)-T)+\delta_2\nabla^2 T \;\le\; \partial_t T \;\le\; \beta\, T\,(1-T)+\delta_2\nabla^2 T, \qquad \vartheta(t):=\alpha\, e^{-\mu(t-t_2)}\downarrow 0.$$

With the logistic supersolution of unit capacity (Lemma 3.3), the upper bound gives $\limsup_t \sup_x T \le 1$. For the lower bound, fix $\varepsilon>0$ and t_ε with $\vartheta(t)\le\varepsilon$ for $t\ge t_\varepsilon$; then $\partial_t T \ge \beta\, T\,(1-\varepsilon-T)+\delta_2\nabla^2 T$, and the *autonomous* logistic subsolution of capacity $1-\varepsilon$ yields $\liminf_t \inf_x T \ge 1-\varepsilon$; ε being arbitrary, $T\to 1$ in $L^\infty(\Omega)$. (The capacity $1-\vartheta(t)$ itself is not a subsolution, since $\frac{d}{dt}(1-\vartheta) = -\vartheta' > 0$.) Finally, $\zeta := H-\gamma/\omega$ solves $\partial_t\zeta = \gamma(T-1)-\omega\zeta+\nabla^2\zeta$, with source $\to 0$ in L^∞ by Step 4. Given $\varepsilon>0$, pick t_3 with $\|\gamma(T(s)-1)\|_{L^\infty}\le\varepsilon$, $s\ge t_3$. The spatially homogeneous functions $\pm(\varepsilon/\omega+\|\zeta(t_3)\|_{L^\infty}e^{-\omega(t-t_3)})$ are super- and subsolutions, so Lemma 3.3 gives $\|\zeta(t)\|_{L^\infty}\le \varepsilon/\omega+\|\zeta(t_3)\|_{L^\infty}e^{-\omega(t-t_3)}$, hence $\limsup_t\|\zeta(t)\|_{L^\infty}\le\varepsilon/\omega$, and $\varepsilon\downarrow 0$ gives $H\to\gamma/\omega$ uniformly. With Steps 3–4, $u(t)\to E_2$ in $L^\infty(\Omega)$. □

S7 The Volterra functional on branch (i)

In this section we construct the Volterra functional on branch (i). Theorem 3.15 settles convergence but says nothing about how it happens. The Volterra functional below, of the form introduced by Goh (1977) and made standard by Korobeinikov and Wake (2002), supplies a quantity that decreases along every solution, on the whole of branch (i), with weights in closed form. Throughout, set

$$\rho := \frac{\alpha\delta}{\delta_c} = \frac{\alpha\delta\gamma}{\omega}, \qquad c_1 = 1, \qquad c_2 = \frac{\rho^{2/3}}{\alpha^2\beta}, \qquad c_3 = \frac{\delta^2}{\omega\,\rho^{2/3}}. \tag{S20}$$

On branch (i), $0<\rho<1$ because $\alpha<1$ and $\delta<\delta_c$, and the weights are positive and finite because $\alpha,\delta>0$; in the notation of Section 3.3, $q=\omega(1-\rho)>0$. Remark S7.3 explains where the weights come from.

We first record the persistence estimate that makes the functional a *bona fide* Lyapunov functional: it keeps N and T uniformly away from zero, so that the functional stays finite and continuous on the absorbing set, and it is where the two defining inequalities of branch (i) are used.

Lemma S7.1 (Uniform persistence on branch (i))**.** *Let $\alpha<1$ and $\delta<\delta_c$, and let u be the solution of* (2.2) *with $u_0\in\mathcal{R}$ and $N_0,T_0\ge c_0>0$ a.e. Then there exist constants $\underline{N},\underline{T}>0$, depending only on the parameters, and a time $t_*=t_*(u_0)$ such that*

$$N(x,t)\ge \underline{N}, \qquad T(x,t)\ge \underline{T}, \qquad \text{for a.e. } x\in\Omega,\ t\ge t_*. \tag{S21}$$

Explicitly one may take $\underline{T}=(1-\alpha)/2$ and $\underline{N}=\frac{1}{2}\big(1-\delta/\delta_c\big)$, both strictly positive exactly because $\alpha<1$ and $\delta<\delta_c$.

Proof. Each step is the scalar comparison Lemma 3.3 with a spatially homogeneous barrier.

Step 1 of the proof of Theorem 3.17 (Section S6.4) uses only $\alpha<1$ and gives $\liminf_t\inf_x T\ge 1-\alpha$, so $T\ge\underline{T}$ for large t.

The acid enters the N-equation with a minus sign; only an upper bound is needed: $T\le 1 \Rightarrow \partial_t H\le\gamma-\omega H+\nabla^2 H$, and the decreasing supersolution $\overline{v}(t)=\gamma/\omega+(\|H_0\|_\infty-\gamma/\omega)^+e^{-\omega t}$ gives $\limsup_t\sup_x H\le\gamma/\omega$, so $H\le\gamma/\omega+\varepsilon$ for large t.

Then $\partial_t N\ge N\big(1-\delta(\gamma/\omega+\varepsilon)-N\big)+\delta_1\nabla^2 N$, with logistic capacity $1-\delta/\delta_c-\delta\varepsilon>0$ for small ε exactly because $\delta<\delta_c$. The comparison needs N bounded away from 0 at some time: on $\mathcal{R}$, $N(1-N-\delta H)\ge -\delta M_0\, N$, so $c_0e^{-\delta M_0 t}$ is a subsolution and $N(x,t)>0$ for finite t (by the uniform bound on N_0; no connectedness or heat-kernel positivity, unlike Theorem 3.16 with $N_0\not\equiv 0$). The logistic comparison then gives $\liminf_t\inf_x N\ge 1-\delta/\delta_c$. □

Theorem S7.2 (Global stability of E_3)**.** *Assume that (α,δ) lies on branch* (i) *of* (3.8)*, i.e. $\alpha<1$ and $\delta<\delta_c$, and let c_1,c_2,c_3 be the weights* (S20)*. Let $u_0\in\mathcal{R}$ with $N_0,T_0\ge c_0>0$ a.e. Then $\mathcal{L}$ of* (S22) *is a strict Lyapunov functional on the absorbing set, and the solution of* (2.2) *satisfies*

$$\big(N(\cdot,t),\,T(\cdot,t),\,H(\cdot,t)\big)\xrightarrow[t\to\infty]{L^2(\Omega)}(N^*,\,T^*,\,H^*).$$

Proof. Define the Volterra-type functional

$$\mathcal{L}(t) = \int_\Omega \Big[c_1 \Big(N - N^* - N^* \ln \tfrac{N}{N^*} \Big) + c_2 \Big(T - T^* - T^* \ln \tfrac{T}{T^*} \Big) + \tfrac{c_3}{2} \big(H - H^* \big)^2 \Big] dx, \tag{S22}$$

≥ 0, $= 0$ exactly at E_3. As $N - N^* - N^* \ln(N/N^*) \to \infty$ when $N \downarrow 0$, finiteness and continuity need Lemma S7.1: for $t \geq t_*$ the orbit lies in

$$\mathcal{R}_* = [\underline{N}, 1] \times [\underline{T}, 1] \times [0, \gamma/\omega + \varepsilon], \qquad \underline{N}, \underline{T} > 0,$$

(the bound $H \leq \gamma/\omega + \varepsilon$, for a fixed $\varepsilon > 0$ and large t, coming from Remark 3.6), where N, T stay uniformly away from 0, $\mathcal{L}$ is finite and L^2-Lipschitz, and $\nabla(N^*/N) \in L^2$ (so the chain rule below is licit). We work on $[t_*, \infty)$.

Step 1 (the reactive part). Let $n = N - N^*$, $\tau = T - T^*$, $h = H - H^*$, $\xi = (n, \tau, h)^\top$. In $\frac{d}{dt}$(S22) the reaction terms give $\int_\Omega [c_1 \frac{n}{N} N(1 - N - \delta H) + c_2 \frac{\tau}{T} \beta T (1 - T - \alpha N) + c_3 h(\gamma T - \omega H)]\, dx$; by $1 - N^* - \delta H^* = 0$, $1 - T^* - \alpha N^* = 0$, $\gamma T^* = \omega H^*$ the brackets become $-n - \delta h$, $-\tau - \alpha n$, $\gamma\tau - \omega h$, so

$$\left. \frac{d\mathcal{L}}{dt} \right|_{\text{react}} = - \int_\Omega \big[c_1 n^2 + c_2 \beta \tau^2 + c_3 \omega h^2 + c_1 \delta\, nh + c_2 \alpha\beta\, \tau n - c_3 \gamma\, \tau h \big] dx = - \int_\Omega \xi^\top Q(c)\, \xi \, dx, \tag{S23}$$

with

$$Q(c) = \begin{pmatrix} c_1 & c_2\alpha\beta/2 & c_1\delta/2 \\ c_2\alpha\beta/2 & c_2\beta & -c_3\gamma/2 \\ c_1\delta/2 & -c_3\gamma/2 & c_3\omega \end{pmatrix}.$$

Step 2 (the diffusive part). The diffusion terms give

$$\int_\Omega \Big[c_1 \tfrac{n}{N}\, \delta_1 \nabla^2 N + c_2 \tfrac{\tau}{T}\, \delta_2 \nabla^2 T + c_3 h\, \nabla^2 H \Big] dx \\ = -c_1\delta_1 N^* \int_\Omega \frac{|\nabla N|^2}{N^2} - c_2\delta_2 T^* \int_\Omega \frac{|\nabla T|^2}{T^2} - c_3 \int_\Omega |\nabla H|^2 \;\leq\; 0,$$

integrating by parts (Neumann: no boundary term), with $\nabla(n/N) = N^* \nabla N / N^2$ (same for T); both integrals converge since $N \geq \underline{N}$, $T \geq \underline{T}$ on $\mathcal{R}_*$ (Lemma S7.1).

Step 3 (Sylvester). The leading principal minors of $Q(c)$ are $m_1 = c_1$, $m_2 = c_1 c_2 \beta - c_2^2 \alpha^2 \beta^2 / 4$ and $m_3 = \det Q(c)$, and, with $c_1 = 1$, expanding the determinant gives

$$\frac{4 \det Q(c)}{c_2 c_3} = 4\beta\omega - \alpha\beta\delta\gamma - \Big[\frac{c_3 \gamma^2}{c_2} + c_2 \alpha^2 \beta^2 \omega + \frac{\beta \delta^2}{c_3} \Big]. \tag{S24}$$

At the weights (S20) each of the three bracketed terms equals $\beta\omega\rho^{2/3}$ (for the first, use $\alpha\delta\gamma = \omega\rho$), and $\alpha\beta\delta\gamma = \beta\omega\rho$. With $r := \rho^{1/3} \in (0, 1)$ this yields

$$m_1 = 1, \qquad m_2 = c_2\beta\Big(1 - \frac{\rho^{2/3}}{4}\Big), \qquad m_3 = \frac{c_2 c_3 \beta\omega}{4} \big(4 - r^3 - 3r^2\big) = \frac{c_2 c_3 \beta\omega}{4} (1 - r)(r + 2)^2,$$

all strictly positive since $r < 1$. By Sylvester's criterion $Q(c) \succ 0$, so $d\mathcal{L}/dt|_{\text{react}} \leq 0$ and, by Step 2, $d\mathcal{L}/dt \leq 0$.

Step 4 (LaSalle). $d\mathcal{L}/dt = 0 \Rightarrow \xi \equiv 0$ a.e. (positive definiteness) and $\nabla N = \nabla T = \nabla H = 0$ a.e., so the largest invariant set in $\{d\mathcal{L}/dt = 0\}$ is $\{E_3\}$. Along this orbit, $u(\cdot, t) \in \mathcal{R}_*$ a.e. for $t \geq t_*$ (Lemma S7.1), a constraint closed in $\mathbf{H}$, so $\omega(u_0) \subseteq \mathcal{R}_*$; there $\mathcal{L}$ is finite, continuous, non-increasing (Steps 1–3) and ≥ 0, and $\{u(t) : t \geq t_*\}$ is precompact in $\mathbf{H}$ (Lemma 3.5). LaSalle's principle for infinite-dimensional semiflows (Henry 1981; Hale 1988) gives $\omega(u_0) \subseteq \{E_3\}$, i.e. $u(t) \to E_3$ in $L^2(\Omega)$. □

Remark S7.3 (On the weights). The weights (S20) are optimal within the Volterra family. In (S24) the product of the three bracketed terms, $\alpha^2\beta^3\gamma^2\delta^2\omega$, does not depend on c, so by the arithmetic-geometric mean inequality the bracket is at least $3\beta\omega\rho^{2/3}$, with equality exactly at (S20). Hence

$$\max_{c>0} \frac{\det Q(c)}{c_1 c_2 c_3} = \beta\omega\Big(1 - \frac{\rho}{4} - \frac{3}{4}\rho^{2/3}\Big),$$

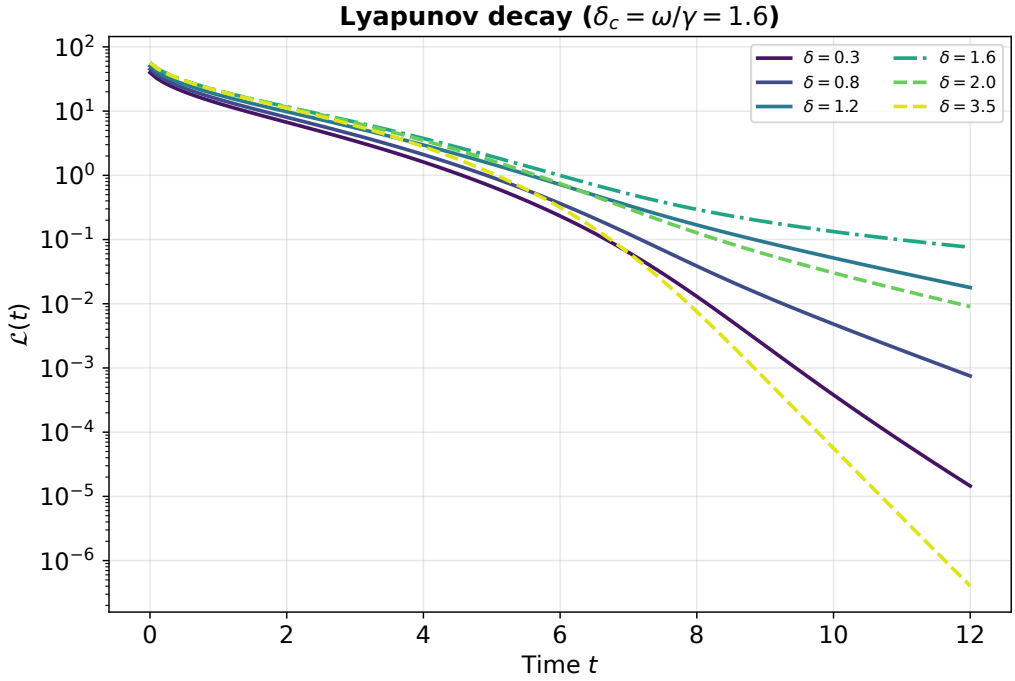


Figure S1: The Volterra functional $\mathcal{L}(t)$ ($\delta_c = 1.6$): solid, $\delta < \delta_c$, relative to E_3; dashed, $\delta > \delta_c$, the proxy $\mathcal{L}_2$ relative to E_2; dash-dotted, the proxy at $\delta = \delta_c$, a non-hyperbolic case covered by no theorem and shown for comparison only.

which is positive if and only if $\rho < 1$: the family certifies global stability exactly where E_3 is locally stable and fails on branch (ii), where $\rho > 1$ and E_3 is a saddle (script, Section 14).

The existence of *some* admissible weights is classical, and we do not claim it. Writing the kinetics at E_3 as $\dot{\xi}_i = u_i(B\xi)_i$ with interaction matrix $B = \left(\begin{smallmatrix} -1 & 0 & -\delta \\ -\alpha\beta & -\beta & 0 \\ 0 & \gamma & -\omega \end{smallmatrix}\right)$ (for H, read $u_3 \equiv 1$), one has $Q(c) = -\frac{1}{2}(CB + B^\top C)$ with $C = \mathrm{diag}(c)$. With $\Sigma = \mathrm{diag}(1, -1, -1)$, the matrix $-\Sigma B \Sigma$ has non-positive off-diagonal entries and leading principal minors 1, β and $\beta\omega(1-\rho)$, so it is a non-singular M-matrix exactly when $\rho < 1$; such a matrix admits a positive diagonal C with $C(-\Sigma B\Sigma) + (-\Sigma B\Sigma)^\top C \succ 0$ (Berman and Plemmons 1994), and Σ commutes with C. What this section adds is the closed form (S20) and its optimality.

At the canonical weights $c_1 = c_2 = c_3 = 1$ the minors are $m_2 = \beta(4 - \alpha^2\beta)/4$ and $m_3 = \Delta/4$, with

$$\Delta := 4\beta\omega - \gamma^2 - \alpha^2\beta^2\omega - \alpha\beta\delta\gamma - \beta\delta^2, \tag{S25}$$

so the unweighted functional is also strictly decreasing when $\alpha^2\beta < 4$ and $\Delta > 0$. That region is strictly smaller than branch (i) and is not invariant under the rescaling of Remark 2.1; in the default regime of Section 4, $\alpha^2\beta = 0.135$ and $\Delta \approx 19.88$, which is why the canonical functional is the one plotted in Figure S1.

S7.1 Numerical decay of the functional

Figure S1 plots, for several δ in branch (i), the functional (S22) at the canonical weights $c_1 = c_2 = c_3 = 1$, strictly decreasing there because $\alpha^2\beta < 4$ and $\Delta > 0$ (Remark S7.3). It decreases monotonically to zero, confirming convergence to E_3. For $\delta \geq \delta_c$, E_3 no longer exists; we plot instead the ad-hoc proxy $\mathcal{L}_2(t) = \frac{1}{2}\int N^2 + \int(T - 1 - \ln T) + \frac{1}{2}\int(H - \gamma/\omega)^2$ (dashed), which is empirically monotone but not proved to decrease. Constructing a quantitative Lyapunov functional for E_2 remains open, whereas convergence itself is established by Theorem 3.15*(ii)*, whose order argument supplies no rate; Theorem 3.17 gives one for N under the stronger condition $\delta(1 - \alpha) > \delta_c$.

S8 The threshold set of the spatial problem

In this section we prove the assertions of Remark 3.13 on

$$\Theta := \partial\mathcal{B}(E_1) \cap \mathcal{U}, \qquad \mathcal{U} := \big\{\, u_0 \in \mathcal{R} \;:\; N_0 \not\equiv 0 \;\text{ and }\; T_0 \not\equiv 0 \,\big\},$$

where $S(t)$ is the semiflow of (2.2) and $\mathcal{B}(E_i)$ is the set of $u_0 \in \mathcal{R}$ whose solution $\to E_i$ in $[L^\infty(\Omega)]^3$. We work in $\mathbf{X}^\theta \hookrightarrow [C(\bar\Omega)]^3$, where $\operatorname{int} K \neq \emptyset$; neighbourhoods and boundaries are relative to $\mathcal{R}$ in that topology.

Mode count. With the Neumann eigenvalues $\mu_{mn} = \pi^2(m^2+n^2)/L^2$ of $\Omega = (0,L)^2$ and $\mu^* \notin \{\mu_{mn}\}$, Proposition 3.11 gives

$$\dim W^u(E_3) = \sum_{\mu_{mn}<\mu^*} 1 \tag{S26}$$

over ordered pairs, $(0,0)$ included. Once $L > L^* := \pi/\sqrt{\mu^*}$ the modes $(1,0)$ and $(0,1)$ are unstable too; at $\mu^* = 5.357$ and $L = 2$ the pairs $(0,0)$, $(1,0)$, $(0,1)$ and $(1,1)$ lie below μ^*, and the count is 4. In the reflection-symmetric subspace only the modes with m, n even remain, and E_3 keeps a single unstable direction there while $L < 2\pi/\sqrt{\mu^*}$.

Codimension of the stable set of E_3. The global stable set, $\bigcup_n$ of the $S(n)$-preimages of the local one, has the same codimension. If $W^s_{\mathrm{loc}} = g^{-1}(0)$ for a submersion g onto $\mathbb{R}^4$ near E_3, the preimage is $(g\circ S(n))^{-1}(0)$, and $Dg\circ DS(n)$ is onto $\mathbb{R}^4$: $DS(n)$ has dense range in $\mathbf{X}^\theta$ (Henry 1981, §3.3, §6.1), and a continuous surjection onto a finite-dimensional space maps any dense subspace onto it (compactness of $DS(n)$ would not suffice, its range not being closed).

Strong monotonicity, precompactness, upward closure. $\mathcal{U}$ is relatively open (on non-negative data its conditions read $\|N_0\|_\infty > 0$, $\|T_0\|_\infty > 0$), positively invariant by the strong maximum principle, and the semiflow is strongly monotone on it. This restriction is harmless: $N_0 \equiv 0$ keeps the orbit on the invariant face $\{N \equiv 0\}$, where T is logistic, H follows, and the limit is E_2; symmetrically $T_0 \equiv 0$ gives E_1, and both $\equiv 0$, E_0.

Precompactness in $\mathbf{X}^\theta$ (not only in $\mathbf{H}$ as in Lemma 3.5) follows from parabolic smoothing: solutions stay in $\mathcal{R}$, so $\|F(u(s))\|_{\mathbf{H}}$ is bounded on every orbit, and variation of constants from $t-1$, with $\|e^{\tau D\Delta_N}\varphi\|_{\mathbf{X}^{\theta'}} \le C\tau^{-\theta'}\|\varphi\|_{\mathbf{H}}$ for $\theta' \in (\theta,1)$ and $\tau \in (0,1]$ of Henry (1981, Thm. 1.4.3) (singularity integrable at $\tau = 0$), bounds $\|u(t)\|_{\mathbf{X}^{\theta'}}$ uniformly in $t \ge 1$ and $u_0 \in \mathcal{R}$; by compactness of $\mathbf{X}^{\theta'} \hookrightarrow \mathbf{X}^\theta$, $\{S(t)u_0 : t \ge 1,\ u_0 \in \mathcal{R}\}$ is precompact in $\mathbf{X}^\theta$. So convergence in $[L^\infty(\Omega)]^3$ upgrades to $\mathbf{X}^\theta$, and $\mathcal{B}(E_1)$ is relatively open, as the union of the $S(t)$-preimages of a neighbourhood of E_1 contained in it, which exists since $J(E_1) - \mu D$ has negative spectral bound and linearised stability (Henry 1981, §5.1) applies.

Finally, $\mathcal{B}(E_1)$ is closed upwards in $\mathcal{R}$: $z \in \mathcal{B}(E_1)$, $z \le_K w \in \mathcal{R}$, and $E_1 = \max_{\le_K}\mathcal{R}$ give, by monotonicity, $S(t)z \le_K S(t)w \le_K E_1$, hence $\|S(t)w - E_1\|_\infty \le \|S(t)z - E_1\|_\infty \to 0$.

The two properties of Θ. They are standard for strongly monotone semiflows (Hirsch 1988; Takáč 1991; Smith 1995), under hypotheses we have not checked; we prove the restricted forms needed.

Θ is invariant and contains no two K-comparable points. Invariance: $\mathcal{U}$ is positively invariant, $S(t)x \in \mathcal{B}(E_1) \Rightarrow x \in \mathcal{B}(E_1)$, and $S(t)$ is continuous. Suppose $x <_K y$ in Θ. Strong monotonicity gives $S(1)x \ll_K S(1)y$, both in Θ. As $S(1)x \in \partial\mathcal{B}(E_1)$ and $\operatorname{int} K$ is open, $\exists\, z \in \mathcal{B}(E_1)$ close enough to $S(1)x$ that $S(1)y - z \in \operatorname{int} K$; upward closure puts $S(1)y$ in $\mathcal{B}(E_1)$, a contradiction.

Θ is a Lipschitz graph over its projection onto any closed complement of a strictly positive direction of K. $e = (1,-1,-1)$ is a constant in $\mathbf{X}^\theta$, interior to K in $[C(\bar\Omega)]^3$, so $\exists\, r > 0$: $e + \varphi \in K$ for $\|\varphi\|_{\mathbf{X}^\theta} < r$. Split $\mathbf{X}^\theta = V \oplus \mathbb{R}e$, V closed, $x = Px + \lambda(x)e$. For $x \neq y$ in Θ, $|\lambda(x-y)| > \|P(x-y)\|/r$ would give $x - y \in K \cup (-K)$, forbidden above; hence $\|x-y\| \le (1 + \|e\|/r)\,\|P(x-y)\|$, and Θ is the graph of a Lipschitz map $P(\Theta) \subset V \to \mathbb{R}e$. Codimension one would further require $P(\Theta)$ to be open in V, which we do not prove.

Irreducibility at a non-constant steady state. Let $u^\sharp = (N^\sharp, T^\sharp, H^\sharp)$ be a non-constant steady state lying in Θ; we do not prove that one exists. Instability follows from Kishimoto–Weinberger (1985), given an irreducible linearisation. As $\Theta \subset \mathcal{U}$, $N^\sharp, T^\sharp \not\equiv 0$, and the strong maximum principle on the connected domain Ω gives $N^\sharp, T^\sharp > 0$; the couplings $\delta N^\sharp$, $\alpha\beta T^\sharp$ and γ are then non-zero and the linearisation is irreducible.

Table S2: Worst pointwise violation of $\mathcal{R}$, over all steps and all six bounds, for each experiment. No clipping is applied.

Experiment	**Section**	**Worst violation**
Convergence study (all N_x, Δt)	4.2	0 (exact)
Bifurcation diagram	4.3	0 (exact)
Regime overlay	4.4	0 (exact)
Macroscopic and sup-norm runs	4.4	0 (exact)
Bistability, small / large seed	4.5	$1.3 \cdot 10^{-15}$
Fisher–KPP fronts, every mesh and every window, including the long-time control to $t = 250$	4.6	$2.9 \cdot 10^{-13}$
Critical-radius sweeps (bisections, all boxes)	4.5	$< 10^{-14}$

We do not claim $\Theta = \partial\mathcal{B}(E_2) \cap \mathcal{U}$ inside the order interval: this would need $\mathcal{B}(E_1) \cup \mathcal{B}(E_2)$ dense there, i.e. a third attractor with non-empty basin ruled out, which we cannot prove. That non-constant steady states exist in Θ for $L > L^*$ is suggested by the measurements of Section S11.1 but not proved, and whether Θ bounds the basin of E_2 and is the stable set of a single critical state is open (compare the scalar case, Du and Matano 2010).

S9 Discrete invariance and convergence of the scheme

Two properties of the scheme are checked here. The first is the discrete counterpart of the invariance of $\mathcal{R}$, which on the square used here is given by Lemma 3.2, since the boundary is Lipschitz but not $C^{2,\nu}$. At every step we record the worst pointwise violation of each of the six bounds, and Table S2 reports the maximum over the run for each numerical experiment in the paper. All values lie at the level of floating-point round-off, so the fully discrete operator inherits the invariance of $\mathcal{R}$, as the maximum principle for the backward-Euler diffusion step and the inward-pointing reaction field lead one to expect.

The second is convergence under refinement. Figure S2 reports pairwise discrepancies $\|u_h - u_{h/2}\|_{L^\infty}$ and $\|u_{\Delta t} - u_{\Delta t/2}\|_{L^\infty}$ at $t = 4$ for the moderate phenotype, over four successive refinements in each direction: $N_x \in \{51, 101, 201, 401, 801\}$ at fixed $\Delta t = 2.5 \cdot 10^{-3}$, and $\Delta t \in \{2 \cdot 10^{-2}, 10^{-2}, 5 \cdot 10^{-3}, 2.5 \cdot 10^{-3}, 1.25 \cdot 10^{-3}\}$ at fixed $N_x = 101$. The number of intervals doubles at each step ($N_x - 1 = 2^k \cdot 50$), so the coarse nodes are *exactly* a subset of the fine ones; $N_x = 50, 100, 200$ would not nest, and the comparison would evaluate the two fields at different points. The observed orders are 1.969, 1.994 and 1.998 under mesh refinement, and 0.997, 0.999 and 0.999 under time refinement. At *fixed* Δt the temporal error cancels in the difference, so panel (a) isolates the spatial error. The measured order is the design order, and it is the boundary treatment of Section 2.4 that secures it. Repeating the same study with the halved boundary rows $(u_1 - u_0)/h^2$, the form appropriate to a cell-centred grid, returns errors of $1.9 \cdot 10^{-2}$, $8.9 \cdot 10^{-3}$ and $4.4 \cdot 10^{-3}$ against $1.6 \cdot 10^{-3}$, $4.1 \cdot 10^{-4}$ and $1.0 \cdot 10^{-4}$, and observed orders 1.11 and 1.03 against 1.97 and 1.99: the boundary layer dominates the L^∞ norm whatever the accuracy of the interior stencil. Panel (b), at fixed N_x, shows first order, inherited from the backward-Euler diffusion step.

S10 Measurement of the front speed

A fit on an early window returns $c_{\mathrm{num}}/c^* \approx 0.93$, and the artefact comes from the window, not the mesh: refining from three to twenty-three nodes across the wing, whose width is $\sqrt{\delta_2/\beta} \approx 0.058$, leaves the ratio at 0.9322, 0.9300, 0.9297 and 0.9296 ($N_x = 201$ to 1601). A pulled front approaches its speed only algebraically, $x_{\mathrm{front}}(t) = c^* t - \frac{3}{2\lambda^*} \ln t + O(1)$ with $\lambda^* = \sqrt{r/\delta_2} \approx 14.5$ (Bramson 1983), so any early fit underestimates c^* by an amount decaying like $1/t$.

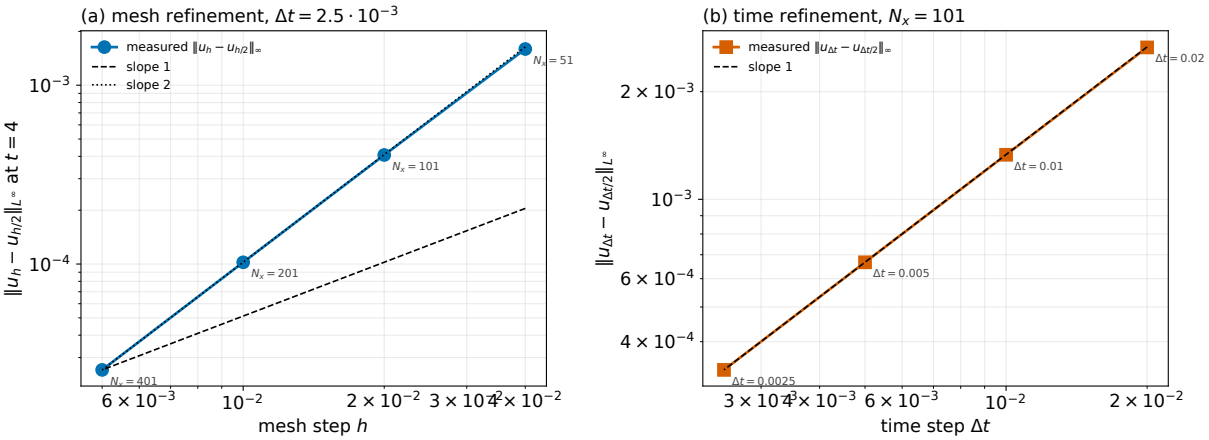


Figure S2: Convergence of the IMEX scheme at $t = 4$ (moderate phenotype): **(a)** mesh refinement, second order; **(b)** time refinement, first order. `main.py`.

With $h = 5 \cdot 10^{-3}$ (about twelve nodes across the wing), the ratio measured on successive late windows climbs from 0.978 on $[10, 20]$ through 1 to 1.004 on $[55, 75]$ (Figure 7(c)). The overshoot belongs to the front itself. Halving h and Δt leaves it unchanged ($1.0037 \to 1.0036$); the Bramson tail is a deficit and can only lower the measurement; and the overshoot is not transient: when the run is continued to $t = 250$ on $[0, 42] \times [0, 0.1]$, the ratio rises and saturates at 1.0037, 1.0046, 1.0050, 1.0052, 1.0053 and 1.0053 on six windows from $[55, 75]$ to $[210, 250]$, the first of which reproduces the shorter-domain value. The asymptotic speed exceeds c^* by 0.53 %.

S11 The size threshold: controls and further measurements

These are the controls behind the size threshold of Section 4.5. The critical radius is obtained by bisection on the seed radius, and the fate of each run is decided locally from the tumour peak and mass (`critical_radius.py`).

Refinement. On $[0, 2]^2$ the bisection returns $R_c = 0.582 \pm 0.002$; refining the mesh from $h = 0.02$ to 0.0125 moves it to 0.579, a shift of the order of the bracket, and halving the time step leaves it unchanged. The domain sweep uses $h = 0.02$ up to $L = 4$ and 0.03 beyond, and the two meshes agree at $L = 4$. In the ω sweep, the least resolved point is $\omega = 32$, where halving h moves R_c from 0.391 to 0.393. The two points added at $\omega = 64$ and 128 use $h = 0.01$ and 0.0075, about twelve nodes per ℓ_H; at $\omega = 128$ the acid relaxes on $1/\omega < 0.008$, comparable to $\Delta t = 5 \cdot 10^{-3}$, and halving Δt gives $R_c = 0.2049$ against 0.2049.

Robustness of the verdict. The invasion test is triggered when the tumour mass has grown by a quarter from its running minimum; near-critical foci that are eventually cleared rise only to 1.10 and 1.07 times that minimum on the plateau boxes. Bisection converges to *a* change of outcome, so we test monotonicity: fifteen radii from 0.20 to 0.90 on $[0, 2]^2$ and nine from 0.35 to 1.15 on the plateau box each give a single transition, which brackets the bisected value.

Saturation of $R_c(L)$. The increment $\mathrm{d}R_c/\mathrm{d}L$ falls from 0.30 on $[1.2, 1.6]$ to 0.05 on $[3, 4]$ and to zero on $[4, 6]$, the last two thresholds differing by less than 0.006. A saturating fit puts the asymptote at 0.844, 3 % above the largest measured value; this gap is the accuracy of the multiple 2.31, and the main text quotes the measured value. Confinement and an off-centre seed both help the tumour.

Dependence on ω. At fixed δ/δ_c, L/ℓ_H and cell diffusion length, the radius moves by a factor of 2.7 from $\omega = 4$ to 32 while R_c/ℓ_H moves by 6 %; the meshes keep about twelve nodes per ℓ_H. The cell diffusivities, varied over two decades, move the critical radius by less than 4 %; the last of these runs is at the limit of resolution (cell front width 0.018 against a mesh step 0.02), and the biological ratio $\tilde{D}_2/\tilde{D}_3$ lies below the range tested.

S11.1 Further measurements

The plateau exceeds the frozen-seed estimate (4.3) by 28 %. The estimate only requires the acid to become lethal at the *centre* of the seed, and it locates that transition correctly: the centre reaches $\delta H = 1$ between $R = 0.60$ and 0.70, and $1.81\,\ell_H = 0.640$ lies in between. The focus, however, is still cleared at $R = 0.70$ and 0.80 and takes over only at 0.85. Making the centre lethal is necessary but not sufficient for the focus to spread. At equal mass, the seed profile hardly matters: a square and a Gaussian give critical equivalent radii 0.585 and 0.621 against 0.582, and part of the difference comes from the grid.

The threshold state itself is probed by enlarging the box across $L = 2\pi/\sqrt{\mu^*} = 2.715$, where the first reflection-symmetric mode at E_3, $(2, 0)$, becomes unstable (Remark 3.13; the centred discs stay in that symmetric subspace). While the near-critical trajectory lingers, the tumour mass retained is 0.235, 0.219, 0.195 and 0.161 for $L = 2$, 2.5, 3 and 3.5, against $T^* = 0.229$, and the spatial standard deviation, read on two runs per box on either side of the bracket, is 0.15–0.33, 0.40–0.64, 0.61–0.93 and 0.71–1.02 times T^*. Below the predicted size, the threshold state stays close to the constant saddle; above it, the state departs from uniformity by an amount comparable to T^*. The crossing is gradual, as expected from a single mode with a small growth rate over a finite dwell time, and we read it as consistent with Remark 3.13, not as a determination of L^*.

S12 Symbolic verifications (SymPy)

The script `sympy_verification.py` checks every algebraic identity on which a proof rests, each with zero residual; we list what each part verifies, recording only the formulas stated nowhere else. The Routh–Hurwitz coefficients at E_3 and the closed form of the margin (3.13) are verified as identities, together with the margin at every mode (Section S4) and the vanishing of the three two-cycle products of J_3. The Sylvester minors of the Volterra matrix at the weights (S20) and at $c_i = 1$, and the optimality of (S20), are checked as identities (Section S7). On Γ_δ, at E_2 with $\delta = \delta_c$, the null vectors $v = (-\omega/(\alpha\gamma), \omega/\gamma, 1)^\top$ and $w = (1, 0, 0)^\top$ give the Sotomayor coefficients $w^\top F_\delta = 0$, $w^\top (DF)_\delta v = 1/\alpha$ and $w^\top D^2F(v, v) = 2\omega^2(\alpha - 1)/(\alpha^2\gamma^2)$, the last vanishing exactly at $\alpha = 1$; those on Γ_α are given in Section S5.2. Section 11 verifies the segment of equilibria at P^*, its transverse spectrum and the reduced field (3.22) (Section S5.3); Section 12, the modal count at the coexistence saddle on $(0, 2)^2$ (four unstable modes, as in Remark 3.13). Section 13 verifies the cooperative sign pattern of $\Sigma J \Sigma$ and the inward field on the faces of the open box, then integrates the kinetics from eight random initial data for each of 120 random parameter vectors in each monostable region; the largest distances to the predicted limit are $1.4 \cdot 10^{-14}$, $5.3 \cdot 10^{-15}$ and $4.5 \cdot 10^{-16}$ (the canonical weights fail on 83 of the 120 branch-(i) vectors, all certified by the weights (S20)). Section 15 verifies the quasi-static acid profile (4.3), whose continuity at $r = R$ reduces to the Wronskian relation $I_0(z)K_1(z) + I_1(z)K_0(z) = 1/z$, checked numerically to $7 \cdot 10^{-15}$; the bisection of $\delta H(0) = 1$ returns $R_c = 0.6396 = 1.809\,\ell_H$.

References

Alt, Hans Wilhelm and Stephan Luckhaus (1983). "Quasilinear elliptic-parabolic differential equations". In: *Mathematische Zeitschrift* 183.3, pp. 311–341. DOI: 10.1007/BF01176474.

Aubin, J.-P. (1963). "Un théorème de compacité". In: *Comptes Rendus de l'Académie des Sciences* 256, pp. 5042–5044.

Berman, Abraham and Robert J. Plemmons (1994). *Nonnegative Matrices in the Mathematical Sciences*. Vol. 9. Classics in Applied Mathematics. Philadelphia: SIAM. DOI: 10.1137/1.9781611971262.

Bramson, Maury (1983). "Convergence of solutions of the Kolmogorov equation to travelling waves". In: *Memoirs of the American Mathematical Society* 44.285. DOI: 10.1090/memo/0285.

Chueh, K. N., C. C. Conley, and J. A. Smoller (1977). "Positively invariant regions for systems of nonlinear diffusion equations". In: *Indiana University Mathematics Journal* 26.2, pp. 373–392. DOI: 10.1512/iumj.1977.26.26029.

Du, Yihong and Hiroshi Matano (2010). "Convergence and sharp thresholds for propagation in nonlinear diffusion problems". In: *Journal of the European Mathematical Society* 12.2, pp. 279–312. DOI: 10.4171/JEMS/198.

Evans, L. C. (2010). *Partial Differential Equations*. 2nd. American Mathematical Society. DOI: 10.1090/gsm/019.

Fenichel, Neil (1971). "Persistence and smoothness of invariant manifolds for flows". In: *Indiana University Mathematics Journal* 21.3, pp. 193–226. DOI: 10.1512/iumj.1972.21.21017.

Gatenby, R. A. and E. T. Gawlinski (1996). "A reaction-diffusion model of cancer invasion". In: *Cancer Research* 56.24, pp. 5745–5753.

Gatenby, R. A. et al. (2006). "Acid-mediated tumor invasion: a multidisciplinary study". In: *Cancer Research* 66.10, pp. 5216–5223. DOI: 10.1158/0008-5472.CAN-05-4193.

Goh, B. S. (1977). "Global stability in many-species systems". In: *The American Naturalist* 111.977, pp. 135–143. DOI: 10.1086/283144.

Grisvard, Pierre (2011). *Elliptic Problems in Nonsmooth Domains*. Vol. 69. Classics in Applied Mathematics. Reprint of the 1985 Pitman edition. Philadelphia: SIAM. DOI: 10.1137/1.9781611972030.

Gyrya, Pavel and Laurent Saloff-Coste (2011). *Neumann and Dirichlet Heat Kernels in Inner Uniform Domains*. Vol. 336. Astérisque. Société Mathématique de France. DOI: 10.24033/ast.899.

Hale, J. K. (1988). *Asymptotic Behavior of Dissipative Systems*. Vol. 25. Mathematical Surveys and Monographs. American Mathematical Society. DOI: 10.1090/surv/025.

Henry, D. (1981). *Geometric Theory of Semilinear Parabolic Equations*. Vol. 840. Lecture Notes in Mathematics. Springer. DOI: 10.1007/BFb0089647.

Hirsch, Morris W. (1988). "Stability and convergence in strongly monotone dynamical systems". In: *Journal für die reine und angewandte Mathematik* 383, pp. 1–53. DOI: 10.1515/crll.1988.383.1.

Hirsch, Morris W., Charles C. Pugh, and Michael Shub (1977). *Invariant Manifolds*. Vol. 583. Lecture Notes in Mathematics. Springer. DOI: 10.1007/BFb0092042.

Kishimoto, Kazuo and Hans F. Weinberger (1985). "The spatial homogeneity of stable equilibria of some reaction–diffusion systems on convex domains". In: *Journal of Differential Equations* 58.1, pp. 15–21. DOI: 10.1016/0022-0396(85)90020-8.

Korobeinikov, A. and G. C. Wake (2002). "Lyapunov functions and global stability for SIR, SIRS, and SIS epidemiological models". In: *Applied Mathematics Letters* 15.8, pp. 955–960. DOI: 10.1016/S0893-9659(02)00069-1.

Lions, Jacques-Louis and Enrico Magenes (1972). *Non-homogeneous Boundary Value Problems and Applications*. Vol. 181. Grundlehren der mathematischen Wissenschaften. Springer-Verlag. DOI: 10.1007/978-3-642-65161-8.

Martin, N. K. et al. (2010). "Leaky vessels as a potential source of stromal acidification in tumours". In: *Journal of Theoretical Biology* 267.3, pp. 454–460. DOI: 10.1016/j.jtbi.2010.07.041.

Ouhabaz, El Maati (2005). *Analysis of Heat Equations on Domains*. Vol. 31. London Mathematical Society Monographs Series. Princeton University Press. DOI: 10.1515/9781400826483.

Pao, C. V. (1992). *Nonlinear Parabolic and Elliptic Equations*. Plenum Press. DOI: 10.1007/978-1-4615-3034-3.

Pierre, Michel (2010). "Global existence in reaction-diffusion systems with control of mass: a survey". In: *Milan Journal of Mathematics* 78.2, pp. 417–455. DOI: 10.1007/s00032-010-0133-4.

Simon, J. (1987). "Compact sets in the space $L^p(0,T;B)$". In: *Annali di Matematica Pura ed Applicata* 146.1, pp. 65–96. DOI: 10.1007/BF01762360.

Smith, Hal L. (1995). *Monotone Dynamical Systems: An Introduction to the Theory of Competitive and Cooperative Systems*. Vol. 41. Mathematical Surveys and Monographs. American Mathematical Society. DOI: 10.1090/surv/041.

Smoller, J. (1994). *Shock Waves and Reaction-Diffusion Equations*. 2nd. Springer. DOI: 10.1007/978-1-4612-0873-0.

Takáč, Peter (1991). "Domains of attraction of generic ω-limit sets for strongly monotone semiflows". In: *Zeitschrift für Analysis und ihre Anwendungen* 10.3, pp. 275–317. DOI: 10.4171/ZAA/452.

Wiggins, Stephen (1994). *Normally Hyperbolic Invariant Manifolds in Dynamical Systems*. Vol. 105. Applied Mathematical Sciences. Springer. DOI: 10.1007/978-1-4612-4312-0.